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REVIEW

A Comprehensive Review of Audio Segmentation and Classification: from Rule-Based Methods to YOHO and Deep Learning Approaches

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Abstract

Accurate audio segmentation and classification are essential for radio and broadcast analytics, supporting applications like speech recognition, music retrieval, and advertisement monitoring. Radio streams, however, present challenges such as overlapping events, abrupt transitions, and broadcast noise, requiring real-time processing. Traditional rule-based methods struggle with these complexities, while frame-based machine learning models often need inefficient post-processing. This paper reviews current audio analysis techniques, from early methods like short-time energy analysis and spectral features to classical machine learning models such as Hidden Markov Models and Support Vector Machines. More recently, deep learning approaches including convolutional, recurrent, and transformer-based architectures have significantly advanced the field, with models like YAMNet, Spleeter, and Wav2Vec 2.0. We focus on the YOHO algorithm, which reformulates segmentation as boundary regression using computer vision models, offering improved precision and real-time efficiency. The paper also goes through advanced architectures that integrates source separation, YOHO-style regression, and transformer embeddings to achieve precise, timestamped segmentation at broadcast scale. Finally, we discuss future research directions, including noise-robust models, self-supervised learning, multimodal fusion, and real-time solutions.

1 Introduction

The digitization of media, matched with the rapid rise of radio broadcasting and various web streaming services has obviously expanded the scale and complexity of audio data requiring analysis, organization and indexing [1, 2, 3]. For this reason, automatic segmentation and automatic classification of radio audio have become important tasks, forming a basis for applications such as automatic speech recognition, music retrieval, advertisement detection, compliance checking, archiving of audio content, and recommendation of personalized content [1, 2]. However, analysis of actual radio streams is much more complex than isolated analysis of speech or music corpora [2, 4]. Radio audio is very heterogeneous, not only because of the abrupt and irregular jumps in programs, but also due to the frequent advertisements and program jingles, the overlapping acoustic events (for example, speech layered under background music) and the variable amount of noise and distortion introduced into the signal during live transmission [4, 5]. Working in these real conditions requires not just accuracy, but a solution that is efficient enough for operational settings in large-scale broadcast monitoring systems [1, 3].

The development of audio signal segmentation and classification has mirrored the wider development of audio signal processing [2, 3]. Early developments in this area relied heavily on rule-based strategies that were not only predicated on handcrafted features but also on deterministic thresholds [2, 4, 5]. Simple measures of short-time energy, zero-crossing rate, power spectral density, and rupture detection were combined with spectral descriptors such as Mel-Frequency Cepstral Coefficients, Chroma, wavelets, and timbre coefficients [2, 6–10]. This was also the same time that speaker diarization (i.e., who spoke when) was conceived, as a rule-based task [4, 11, 12]. While these methods were easy to interpret and had low computational cost for small systems,

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they faced challenges in noisy or overlapping events and their deterministic thresholds were immoveable across a number of real broadcast situations [4, 5, 11]. The need to predict across events helped drive the embedding of statistical approaches to machine learning techniques into practice, particularly incorporating Hidden Markov Models or Gaussian Mixture Models for temporal segmentation, and Support Vector Machines, k-Nearest Neighbors, Decision Trees, and ensembles for both speech–music discrimination and event classification [2, 4, 13–15]. These methods were far more robust than deterministic systems and approaches, but they continued to heavily depend on manual feature engineering, which resulted in a poor scaling towards real data diversity in shape and size [1–3].

As demonstrated in Fig. 1, different audio representations (raw waveforms, log-Mel spectrograms, and MFCCs) were examined for transformer-based classification tasks [16–18], highlighting the varied strengths of these representations in sampling and/or capturing frequency, time, and perceptualness. This links back to earlier from the Introduction section where the significance of extracting features (spectral-descriptors, MFCC, chroma, etc.) and the typical approach of manually engineering features in classical approaches are discussed [2, 6–10], echoing the consideration to develop models that focus on the diversity of representations that appear in radio audio [1, 18].

The emergence of deep learning systems represented a pivotal moment [18, 19]. Convolutional Neural Networks provided the possibility of automatically learning hierarchical features via spectrograms [20–22], and Recurrent Neural Networks, Long Short-Term Memory networks, and Gated Recurrent Units enabled modeling temporal dependencies present in audio sequences [13–15]. Hybrid CNN-RNN architectures routinely provided leading performance in many sound classification tasks [23–25].

As shown in Fig. 2, the audio signal (raw waveform or spectral representation, e.g., log-Mel spectrogram) is first taken as input to a feature encoder that eventually projects into embeddings, which are projected through transformer layers that self-attend and ultimately feed-forward to learn long-range temporal dependencies. This construction is aligned with modern deep learning sections of the Introduction and particularly the use of transformers to model long term context to build performance in these tasks for classification and segmentation of audio.

More recently, transformer-based architecture introduced attention mechanisms to model long-range dependencies present across sequences and established new state-of-the-art results in speech recognition, sound event detection, and audio tagging [16–18, 26]. Pretrained models, including YAMNet, trained using the extensive Audio Set corpus [27, 28], Spleeter for source separation [29], Pyannote for speaker diarization [30], and Wav2Vec 2.0 for self-supervised audio embeddings [31–34] have reduced reliance on large annotated datasets and have enabled transfer learning to complete tasks that suit respective contexts (e.g., radio). These developments have changed the possibilities of audio segmentation and classification [1, 18, 35]. Nonetheless, most deep learning systems still work in a frame-based fashion providing dense per-frame predictions that require extensive post-hoc processing of segmenting continuous streams of audio into coherent boundaries [36, 37]. The post-processing step often can lead to additional computational overhead and a less efficient process for real-time broadcast analytics [23, 36, 37].

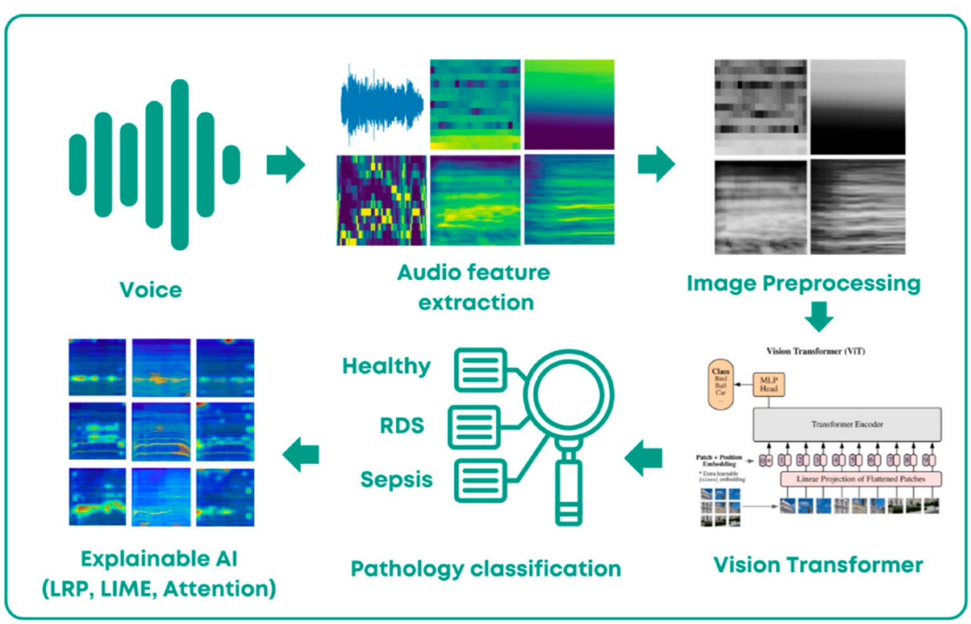


Fig. 1 Comparison of different audio representations (spectrograms, MFCC, waveform) for transformer-based classification, highlighting preprocessing and representation choices

Fig. 2 Transformer-based audio processing pipeline: raw waveform or spectrogram input → embedding encoder → transformer layers with self-attention → classification/segmentation output

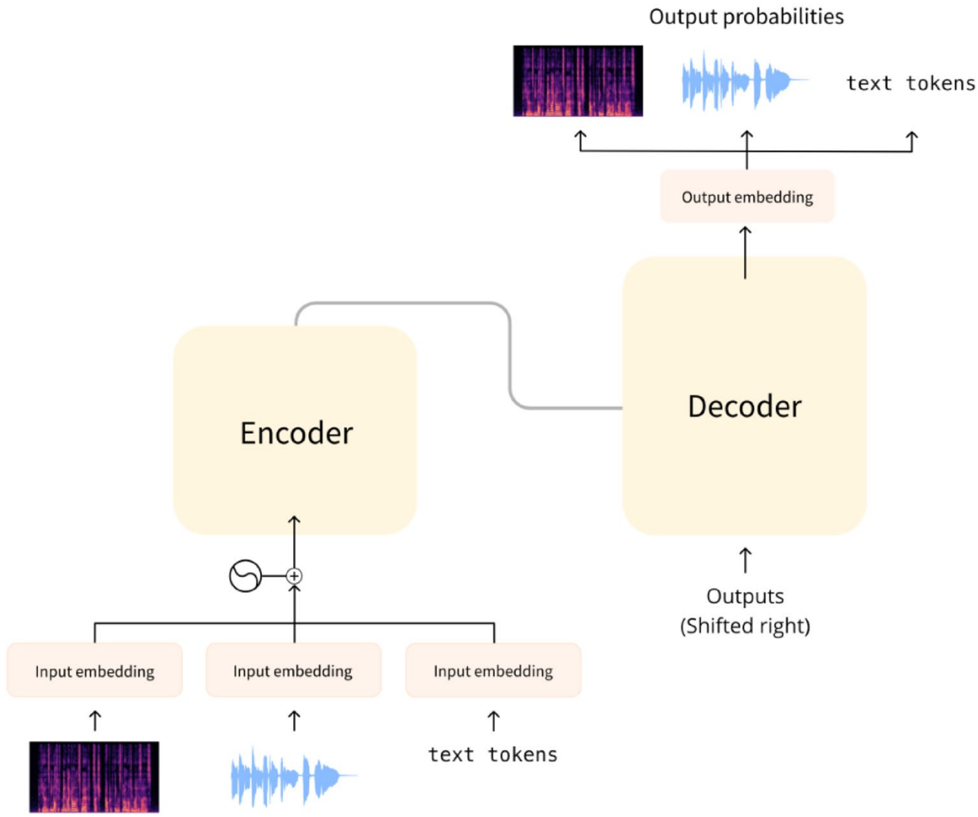
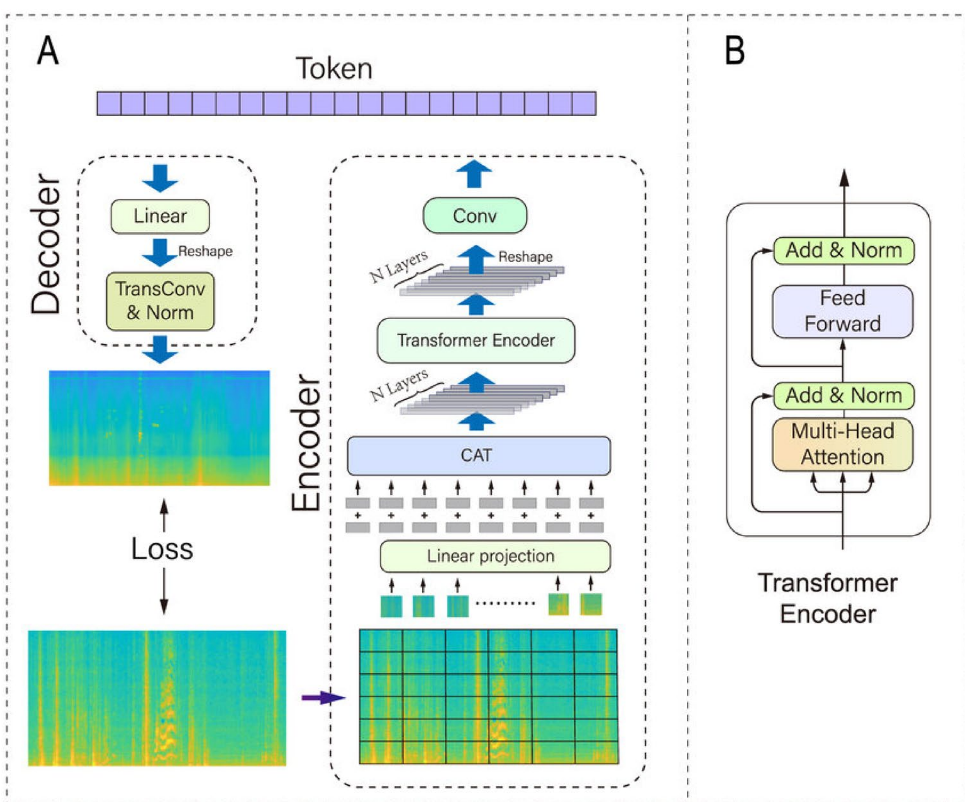


Fig. 3 The architecture of a spectrogram transformer auto-encoder (AST-AE): input spectrogram → encoder layers (self-attention, positional encoding) → latent embedding → decoder / reconstruction output



The AST auto-encoder, as shown in Fig. 3, begins with a spectrogram as input, processes it through multiple transformer encoder layers with positional encoding to keep track of temporal relations, compresses it into a latent embedding, and reconstructs/decodes the representation. This continues the deep learning portion of the Introduction, in which transformer-based models are characterized as allowing for richer, hierarchical feature learning from spectral input, thereby decreasing dependence on hand-crafted features, and improving representation of complicated radio audio that contains overlapping events and noise.

A recent solution to these inefficiencies is the YOHO (You Only Hear Once) algorithm, which reframes segmentation as a boundary regression problem instead of tasks of frame-level classification [38]. Similar innovations in computer vision, like YOLO, inform the YOHO event boundary assessment which can predict the onset and offset boundaries of an event in one step [22, 38, 39]. Critically, YOHO predicts the boundaries without cumbersome post-processing, thereby allowing for both greater temporal precision, and computational efficiency [38]. Lightweight models, such as those built with Mobile Net-based and other efficient backbones improve the value of YOHO for large-scale real-time analysis of radio [38, 40]. Preliminary studies have shown YOHO to meet or exceed the processing speed while preserving the classification accuracy of classical and deep learning algorithms with experiments of independently across paradigms, suggesting that YOHO is a novel use of technology within this task [38].

In spite of these improvements, difficulty will always be present. Radio audio presents difficulty due to its complexity; overlapping acoustical events, noisy settings, class imbalance, and constraints for real-time processing all serve to put even the most sophisticated systems under duress [1, 2, 4, 35]. To add to the complexity, there is a lack of a standard benchmark and evaluations to allow a fair comparison of methods [1–3]. While there are many datasets, this study will use datasets that are common and informative benchmark datasets which have been used, including Audio Set, ESC-50, UrbanSound8K, TUT Acoustic Scenes, GTZAN, MUSAN, and Librispeech, all with at least one limitation regarding either class balance, domain diversity, or annotation quality [27, 28, 34, 41, 42]. To mitigate these challenges, we will need a collection of systems that strive toward balance in robustness, interpretability, computational costs, and replicability toward studies [1, 18, 35].

The workflow shown in Fig. 4 begins with raw audio, proceeds through preprocessing (feature extraction with MFCCs or spectrogram), uses a front-end module (typically CNN-based) to convert to embeddings, then those are fed into transformer encoder layers followed by task-specific output heads (classification, segmentation and/or boundary detection). Fig. D is a reminder of the introduction's discussion of early deep learning approaches that combined CNNs with RNNs, and then more recently how a transformer architecture is being incorporated to replace or augment these, thereby reducing costs and improving performance for diverse tasks involving varied audio events.

The current review addresses this need by providing a comprehensive synthesis of methods that include rule-based, machine learning, deep learning, and regression-based methods, where we discuss YOHO as a leading method [1, 2, 35, 38]. Alongside synthesize existing work, we introduce a hybrid AI-driven Radio Analyzer architecture which combines source separation, YOHO style regression and transformer-based embeddings to deliver accurate and timestamped segmentations and classifications at broadcast scales [16, 17, 26, 29, 30, 34, 38]. Contributions include a unifying taxonomy of approaches, a thorough review of YOHO, a compilation of benchmark datasets and future research roadmap [1–3, 27, 28, 34, 38, 41, 42]. By integrating historical perspectives with analysis of newly emerging trends, we provide a reference for researchers and a practical framework for practitioners developing future radio analytics systems [1, 18, 35].

The recent surveys regarding audio classification using deep learning and transformers for audio detection tasks only classify the audio signals as given in [43]. Unlike these topics, which focuses broadly on deep learning methods for general audio classification, our work specifically addresses:

- Speech segmentation in broadcast streams
- Speaker identification within continuous radio programming
- Time stamp recognition in real-time broadcast settings

Our review emphasizes broadcast radio scenarios, which involve unique challenges such as overlapping speech, music interludes, noise variability, and real-time constraints. While transformers for audio detection tasks focuses primarily on transformer architectures for detection tasks [44], our review provides:

- A task-oriented taxonomy, which starts from segmentation, moves to audio identification, and identifies time-stamp alignment
- Comparative discussion of classical, hybrid, and modern transformer-based pipelines within broadcast applications

In this work, we uniquely analyze and reviewed the integration of Voice activity detection (VAD), Speaker diarization, Speech-to-text alignment, and time stamp refinement within a unified broadcast framework.

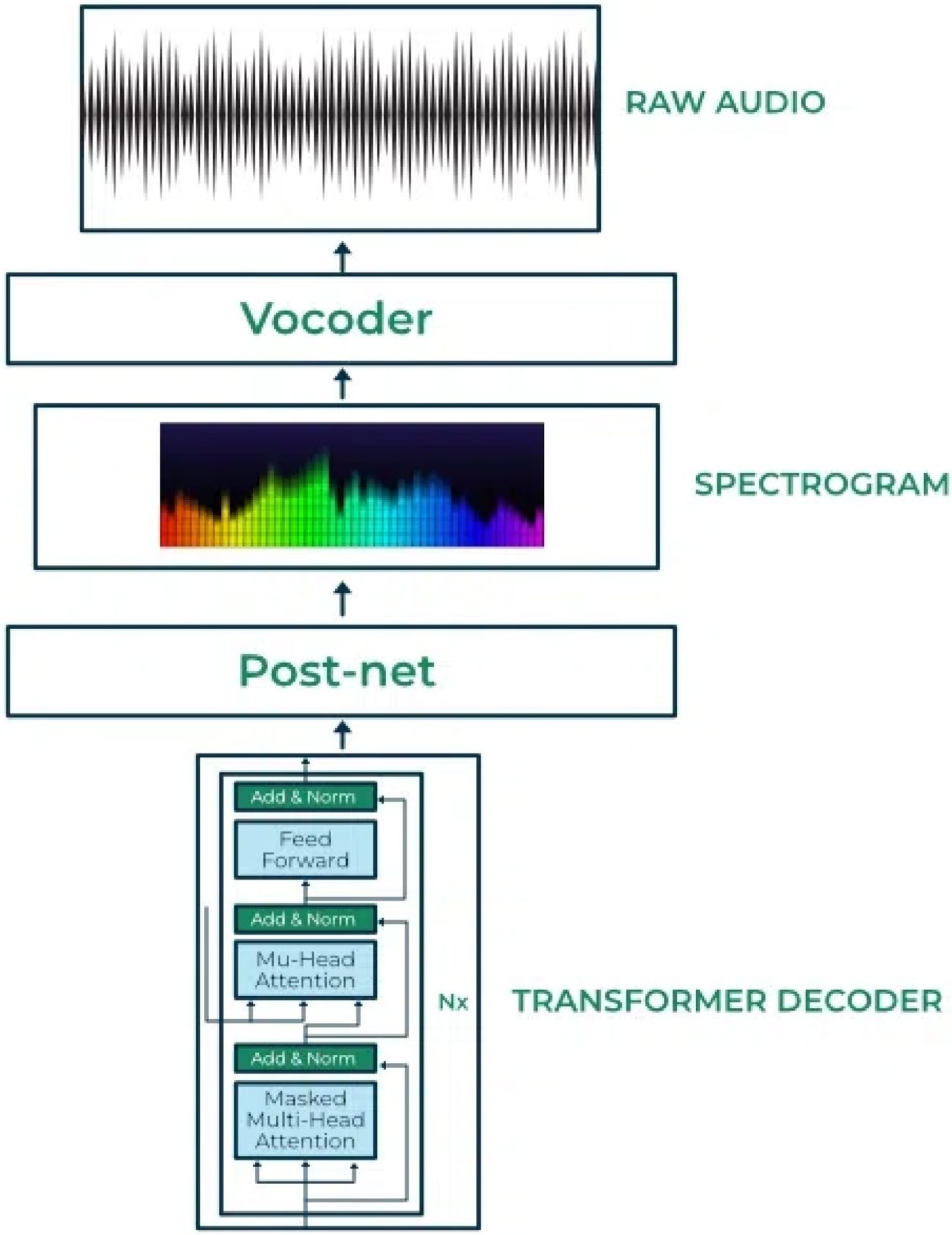


Fig. 4 Feature extraction + transformer encoder workflow: waveform → MFCC/spectrogram → embedding via CNN/front-end → transformer encoder → task heads (classification or boundary regression)

2 Background and Preliminarie

Audio segmentation and classification is a topic that straddles the fields of signal processing, machine learning, and deep learning, and in order to evaluate complex methods, one must first clearly understand several fundamental principles [1–3, 18, 19]. This section describes the foundational technical elements of modern systems: digital audio representation, preprocessing operations, feature extraction methods, fundamental elements of classical models, preliminaries of deep learning, evaluation metrics, and datasets used for standard benchmarking [2, 3, 27, 28, 42, 45]. These elements serve both as the foundational knowledge to support the methods covered later in the review, but as important elements that reveal the challenges of interfacing with radio audio streams [1, 2, 4, 35].

2.1 Digital Audio Representation

The audio signals in the real world are continuous in both time and amplitude. To allow for analysis by a computer they must be converted into a digital representation. This is achieved through sampling and quantization [2, 3, 45].

- **Sampling:** The continuous-time signal $x(t)$ is discretized by taking samples at uniform intervals to produce a discrete-time signal.

$$x[n] = x(nT), \quad \text{where } T = 1/f_s \text{ and } f_s \text{ is the sampling frequency.}$$

  Commonly used frequencies for telephony and coding audio are 16 kHz for speech, 22.05 kHz for music, and 44.1 kHz for compact disc quality sound files [2, 3, 45].
- **Quantization:** Each sample is mapped to one of 2^b discrete values, where b is the bit depth (commonly 16-bit or 24-bit) [2, 45].

Digital audio is generally divided into short overlapping windows (also called frames); each frame is typically 20–40 milliseconds in length. The use of frames allows the audio to be assessed locally in the spectral domain, which is useful when undertaking classification and segmentation applications [2, 6, 45].

As shown in Fig. 5, the preprocessing framework is shown as a modular and flexible system. Raw signals are processed through modules for silence detection, denoising, and amplitude normalization before being processed by the feature extraction portion of the pipeline. The figure makes clear that, preprocessing is not just a single task but typically a coordinated set of tasks which can be adapted for varying and noisy audio environments. By visualizing these elements, the figure further illustrates how preprocessing sets the stage for reliable and proficient downstream modeling.

2.2 Preprocessing Fundamentals

Processing the raw audio signal is standardization, noise-robust not a precondition to the feature extraction [2, 3, 45]. Important steps are:

- **Pre-emphasis filtering:** The high-pass filter in the form of $y[n] = x[n] - \alpha x[n-1]$, boosts the high-frequency

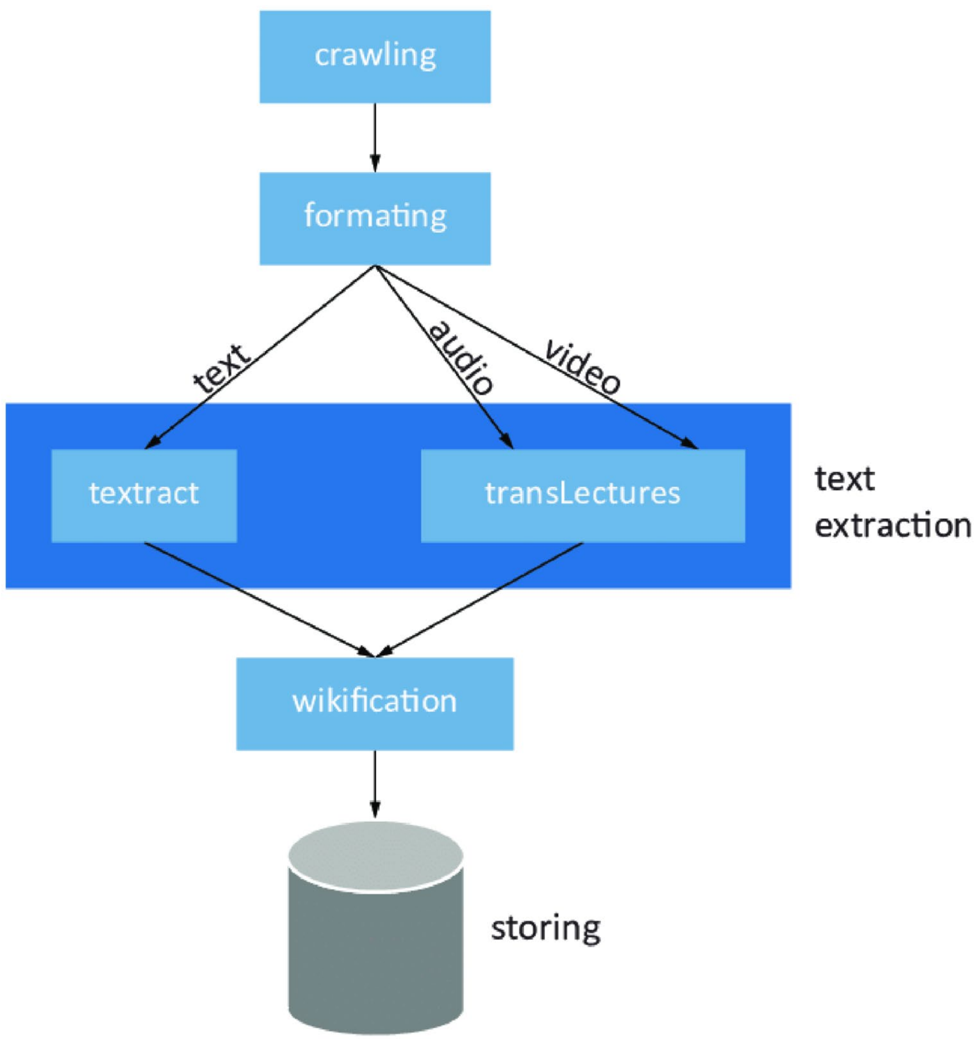


Fig. 5 Modular preprocessing pipeline architecture for raw audio input, showing branching operations for silence removal, noise suppression, and preparation for feature extraction

components of the document, compensating for spectral tilt [2, 3, 45].
- **Framing and windowing:** The signals are split into overlapping frames or segments and are multiplied with window functions (Hamming, Hann, etc.) to reduce spectral leakage [2, 6, 45].
- **Silence removal:** Regions of low-energy are removed to reduce computation [2, 5].
- **Noise reduction:** Classic methods such as Wiener filtering and modern deep denoising autoencoders are used to improve the clarity of the time signal [2, 18, 29].
- **Normalization:** Amplitude normalization makes sure energy levels across recordings are approximately the same [2, 45].

These processes improve robustness and set up the signal for feature extraction [2, 3, 45].

As illustrated in Fig. 6, there is a series of organized preprocessing steps applied to raw audio signals, enabling their input into the system to be standardized and more robust. The figure illustrates how framing and windowing break down the waveform into small overlapping pieces, silence removal removes portions of the audio with low energy, and noise reduction minimizes the negative effects of background distortion. Normalization implies that all of the signals maintain amplitude consistency across recordings, allowing a consistent input into the following stage of feature extraction. This established workflow exemplifies that preprocessing steps are crucial for accurate audio segmentation and classification.

2.3 Feature Extraction Preliminaries

Feature Extraction converts raw waveforms to compact discriminative representations. Features are generally defined based on four categories: temporal, spectral, cepstral, and time–frequency representations [2, 3, 6, 18, 45].

- **Temporal features:** Feature extraction in time may include Short-Time Energy (STE), Zero-Crossing Rate (ZCR), and autocorrelation as measures of the change in a signal over time [2, 5, 45].
- **Spectral features:** Common spectral features that are used in feature extraction include the spectral centroid (perceived brightness), roll-off (the frequency below which 85–95% of the signal's energy is contained), spectral flux (change in the variation over the spectrum), and bandwidth [2, 6–8, 45].
- **Cepstral features:** Mel-Frequency Cepstral Coefficients (MFCCs) are the most commonly used features for audio feature extraction because they model human perception of auditory stimuli. Human perception of auditory stimuli can be modeled using MFCCs, which are arrived at through the steps of DFT to convert the raw waveforms to a frequency domain representation, linear frequency mapping into the Mel scale, logarithm application, and DCT [2, 6, 9, 45].
- **Time–frequency features:** Rich joint time–frequency representations of audio signal can be provided through several techniques including, the Short-Time Fourier Transform (STFT), Mel-spectrograms, constant-Q transform (CQT), and wavelets [2, 3, 6, 10, 17, 18, 45].

The STFT is formally expressed as [2, 3, 6, 18, 45]:

$$X(k, m) = \sum_{n=0}^{N-1} x[n]\, w[n-m]\, e^{-j\frac{2\pi kn}{N}}$$

where $w[n]$ is the window function, m is the frame index, and k is the frequency bin [2, 6, 45].

2.4 Classical Model Foundations

2.4.1 Hidden Markov Models (HMMs):

Hidden Markov models (HMMs) assume that a sequence of hidden states (e.g., speech, music, silence) produces observed features [2, 4, 13–15].

The joint probability of states Q and observations O is:

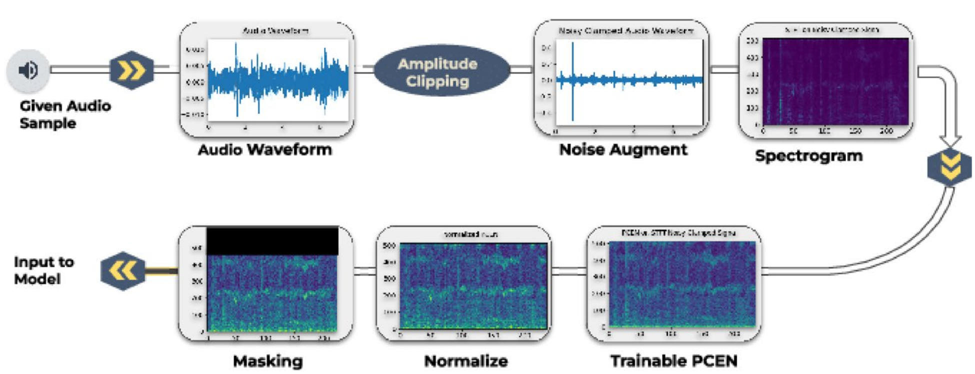


Fig. 6 Data pre-processing pipeline illustrating framing, windowing, silence removal, noise filtering, and normalization before feature extraction

$$P(Q,O) = \pi_{q_1} \prod_{t=2}^{T} a_{q_{t-1}q_t} \prod_{t=1}^{T} b_{q_t}(o_t)$$

where π_{q_1} is the initial probability, a_{ij} are state transition probabilities, and $b_i(o_t)$ are observation likelihoods [2, 4].

2.4.2 Gaussian Mixture Models (GMMs):

GMMs model distributions as a mixture of Gaussians [2, 4]:

$$p(x) = \sum_{i=1}^{M} w_i\, N(x|\mu_i, \Sigma_i)$$

where w_i, μ_i, and Σ_i are weights, means, and covariances. These were widely used for speech–music discrimination [2, 4].

As illustrated in Figure 7, the end-to-end pipeline starts with collecting raw waveforms, which are then transformed by a series of preprocessing steps, including time-frequency transformations (e.g. short-time Fourier transform or Mel-spectrogram) that yield compact, informative representations of the signal. The representations are then passed into learning models to perform tasks such as classification, segmentation, or for event boundary detection. The figure encompasses a view capturing the various stages of the audio analysis pipeline visually, showing how a raw input is transformed into a structured output.

2.5 Deep Learning Preliminaries

Deep learning removes the need for feature engineering by learning direct representations [18, 19].

- **Convolutional Neural Networks (CNNs):** Learn hierarchical spatial features from spectrograms [20–22].
- **Recurrent Neural Networks (RNNs), LSTMs, and GRUs:** Learn long-range temporal dependencies in sequences [13–15].
- **Transformers:** Use self-attention mechanisms for sequence modeling [16, 17, 26]. Attention is computed as:

$$Attention(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where Q, K, and V are query, key, and value matrices [16, 26].

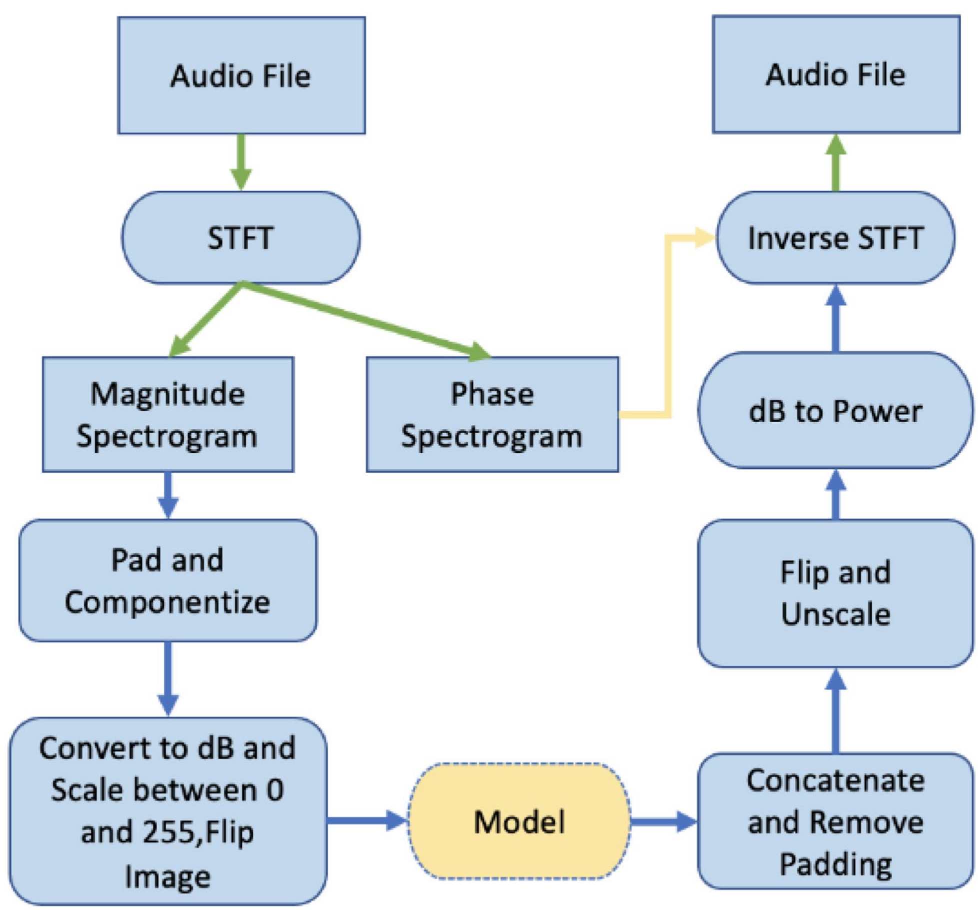


Fig. 7 End-to-end audio processing pipeline, from raw waveform acquisition through preprocessing and spectrogram generation to model-level classification and segmentation

- **Pre-trained models:** Wav2Vec2.0 (self-supervised) [31–33], YAMNet (Audio Set-trained) [27, 28], Spleeter (source separation) [29], and Pyannote (speaker diarization) [30] offer embeddings that transfer across domains.

Figure 8 outlines the full audio processing pipeline using four primary stages. Preprocessing involves preparing the signal, feature analysis involves extracting distinguishable representations, modeling involves applying statistical or deep learning methods, and output captures the final results such as classification labels or identified boundaries. This staged structure emphasizes the layered views of audio analytics, and serves as a conceptual map to illustrate how each component of the system contributes to our overall goal of rapidly and accurately segmenting and classifying audio.

2.6 Evaluation Metrics

Segmentation and classification can be assessed with respect to multiple metrics, such as [1–3, 18, 35]:

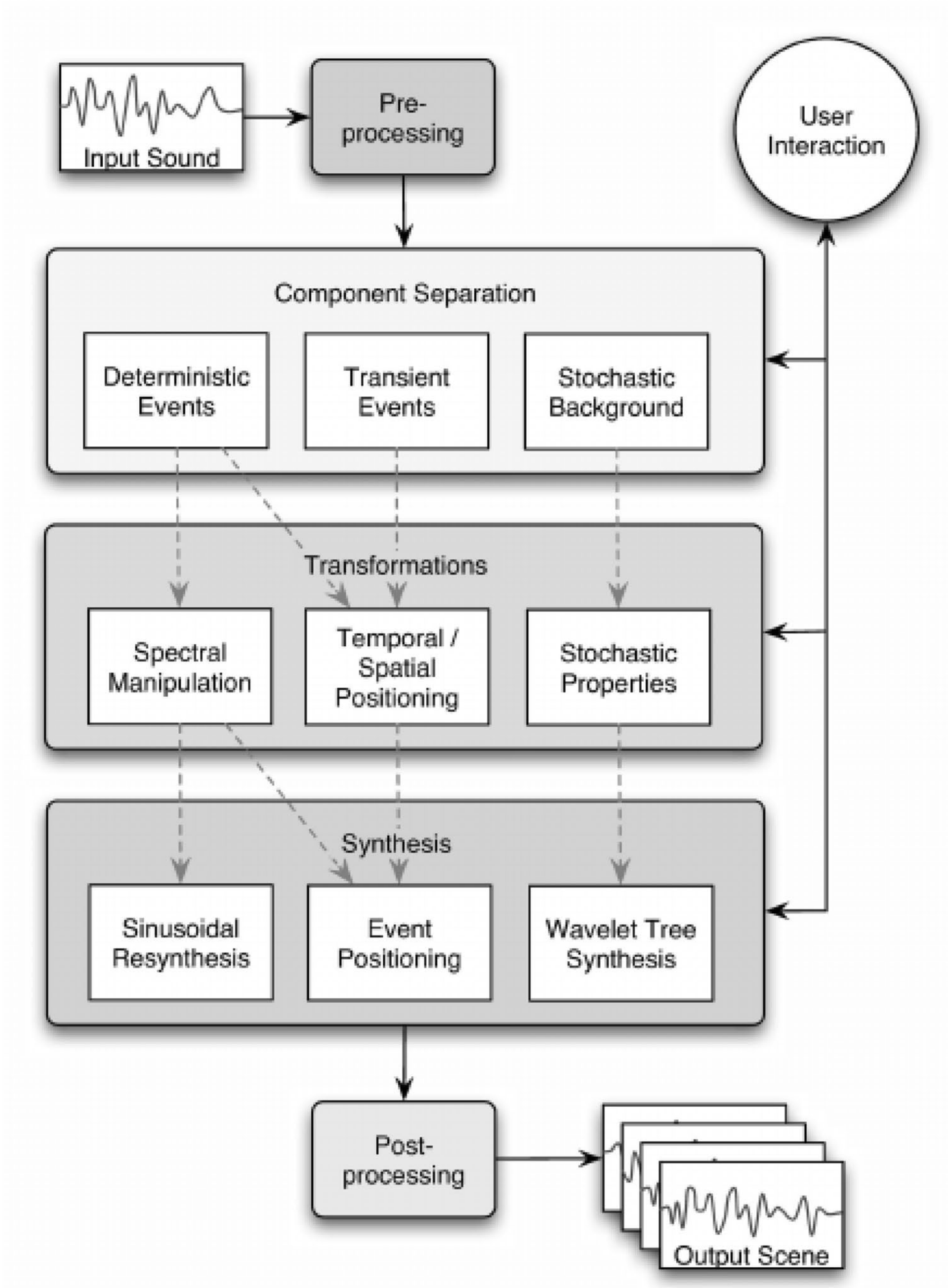


Fig. 8 Stages in an audio pipeline: preprocessing, feature analysis, modeling and transformation, and final system output

- **Accuracy:** The proportion of correct observations relative to the total number of observations, providing an overall measure of model correctness [1, 2].
- **Precision, Recall and F1 score:** These metrics assess the trade-off between false positives (+) and false negatives (-). Precision indicates the accuracy of positive predictions, recall reflects the model's ability to identify all relevant instances, and the F1 score balances both precision and recall, providing a single metric for performance [1, 2, 35].
- **Receiver Operating Characteristic (ROC) and Area Under Curve (AUC):** The ROC curve plots the true positive rate against the false positive rate, while the AUC measures the area under the ROC curve, providing an aggregate measure of performance for binary classification tasks [1, 2, 35].
- **Segmentation Error Rate (SER):** This metric measures how accurately the predicted boundary aligns with the true boundary, indicating the precision of segmentation models in capturing the temporal structure of the audio [36–38].
- **Source-to-Distortion Ratio (SDR):** SDR evaluates the quality of source separation, quantifying how well different sound sources (such as speech and music) are isolated from each other, with higher values indicating better separation quality [18, 29].
- **Latency and throughput:** These metrics are critical for real-time broadcast systems. Latency measures the delay between input and output, while throughput assesses the system's capacity to process data in real-time. Both metrics are essential for ensuring that the system can meet the demands of live broadcasting and time-sensitive applications [1, 35].

2.7 Benchmark Datasets

Datasets of high quality are necessary in order to reproduce research [1–3, 35]. Datasets that have received attention from the research community include:

- **ESC-50:** 50 classes of environmental sounds in a curated and balanced dataset [28].
- **UrbanSound8K:** 8732 short clips across 10 classes of sounds from urban environments [27].
- **AudioSet:** 2,088,320 short clips (10 seconds each) labelled with 527 different sound events [27, 28].
- **TUT Acoustic Scenes:** Datasets representing labelled acoustic environments; a limited number of audio scenes were used [18, 35].
- **LibriSpeech:** A large speech dataset based on audiobooks [34, 41].
- **MUSAN:** A dataset consisting of speech, music, and noise [34, 41].

All datasets have limitations: for instance, ESC-50 and the UrbanSound8K are quite small; everything else is considerably larger, but for example, AudioSet is weakly labelled. In addition to dataset size, dataset imbalance and overlapping classes are other ongoing issues for research purposes with radio audio [1–3, 27, 28, 34, 35, 41].

3 Preprocessing and Feature Extraction

The utility of audio segmentation and classification systems relies heavily on signal preprocessing and feature discriminability [1–3, 18, 45]. Raw audio waveforms reside in an extremely high-dimensional space that may contain extraneous information (redundancy), cavity distortion or transmission distortion, and environmental noise—making direct analysis or transformation impractical [2, 3, 45]. As a result, preprocessing guarantees that signals meet standardization requirements and are noise robust, and a contemporary representation procedure called feature extraction will constrain and organize these analog signals into less descriptive and manageable [compact] representations that will describe aspects of the acoustical information that are necessary for reliable classification [2, 3, 6, 18, 45]. Over time, segmentation and classification have journeyed from simple and non-computational handcrafted descriptors through inferior and easily manipulated representations of audio time, frequency and finally to deep learned embeddings that will serve to be the bedrock of modern audio-based analysis [1, 18, 19, 21].

3.1 Preprocessing Fundamentals

Preprocessing represents the first step, and may actually be the most important step, in segmentation pipelines, as downstream performance of both machine learning and deep learning models highly depends on the quality of the input [2, 18, 45]. Pre-emphasis filtering is considered to be one of the oldest and most commonly used preprocessing techniques, and is simply a high-pass filter that compensates for the natural spectral tilt present in speech and radio recordings [2, 3, 45]. By boosting the higher frequency content, pre-emphasis filtering reinforces the more obscure cue distinctions of speech, music, and environmental sounds [2, 3, 6, 45]. After the pre-emphasis filtering has been implemented, the audio signal is divided into short overlapping frames (typically 20 to 40 ms in duration) because audio signals are quasi-stationary over such short durations [2, 6, 45]. Each frame is subsequently multiplied by a window function (e.g., Hamming or Hann) which reduces spectral leakage during the Fourier transform [2, 6, 45].

$$y[n] = x[n] - \alpha x[n-1], \quad 0.9 \leq \alpha \leq 0.97$$

The removal of silence, or voice activity detection VAD, is another critical aspect of preprocessing [2, 4, 5]. Broadcast recordings will often have extended periods of silence, pauses, or background noise that incur a computational cost without adding useful information; early systems relied on threshold models which accepted or rejected large blocks of audio based on energy detection to remove silence [2, 5]. Recent research has turned to supervised VAD models that are trained as supervised models with large speech corpora [4, 30]. The same audio conditions that may elevate silence will also render noise reduction an important preprocessing step in radio audio, particularly in live broadcast instances, where environmental sounds and distortion within the transmission impact the noise conditions [1, 2, 4, 35]. Classical methods of noise reduction, such as spectral subtraction and Wiener filtering, are still used due to their simplicity, though deep denoising autoencoders and spectral masking techniques are more recent alternatives that have shown advantages in the unpredictable noise conditions present in radio audio [2, 18, 29]. Finally, normalization, and mean–variance scaling can be used to overcome the dynamic range variations created, to be consistent across recordings that are made in varying conditions [2, 6, 45]. These preprocessing tasks, although simple, contribute as foundational to feature extraction and serve to enhance robustness and reproducibility [1, 2, 18, 35].

$$x_m[n] = x[n + mH] \cdot w[n], \quad 0 \leq n < N$$

The raw audio waveform, as depicted in Fig. 9, will undergo a structured pre-processing and feature extraction pipeline. The first step creates frames and windows and subsequently computes spectral content via FFT or similar transforms. After that, we apply a Mel-scaled filterbank to accentuate perceptually important frequency locations, then compute log amplitude in order to arrive at a more compact representation of the spectrum. This pipeline incorporates the process outlined that the audio signal is standardized and robust to noise before being projected to discriminative features that serve as input to a segmentation or classification model.

3.2 Classical Feature Extraction

The initial segmentation utilized hand-crafted descriptors which encapsulated temporal and spectral characteristics [2, 3, 6, 45]. The short-time energy (STE), which is defined as

$$ZCR_m = \frac{1}{2N} \sum_{n=1}^{N} |\mathrm{sgn}(x_m[n]) - \mathrm{sgn}(x_m[n-1])|$$

represents the amplitude of the signal and is rightly small if it is not part of speech, making it effective for detecting speech–silence. The zero-crossing rate (ZCR) captures frequency content [2, 6, 45].

Spectral descriptors obtained from the Discrete Fourier Transform (DFT) convey characteristics including centroid, roll-off, flux, and bandwidth, and can be interpreted as categorical signals containing tonal or broadband information [2, 6–8, 45].

Mel-Frequency Cepstral Coefficients (MFCCs) became the dominant cepstral feature, in part, due to their perceptual basis [2, 3, 6, 45]. The MFCC features pipelined origin is placing the Fast Fourier Transform (FFT) in the signal processing, and then mapping the signal's frequency to the Mel scale.

$$M(f) = 2595 \log_{10}\left(1 + \frac{f}{700}\right),$$

through the Discrete Cosine Transform (DCT) model. Wavelet transforms provide an even broader means of signal representation by being able to apply multi-resolution analysis, which identifies both transient and steady-state events [2, 6, 45].

3.3 Time–Frequency Representations

Although classical features present certain characteristics of audio, they usually do not capture the joint dynamics of the spectral content over time, which is an important trait for segmentation. Time-frequency methods do provide such a representation, as they are a way to visualize spectral energy extent across time frames, are two-dimensional in nature. The Short-Time Fourier Transform (STFT) is a traditional form of time-frequency method, which decomposes the signal in small time frames, with localized frequency components contained in frequency bins [2, 3, 18, 45]. By taking the STFT to produce spectrograms of the audio, we get a view of some patterns such as harmonic structures and transient bursts that assist us in distinguishing speech, music, and noise from one another. Amplitude (magnitude) of the frequency information within frequency bins is the most common time-frequency representation used, although there are variants (for example, the mel-spectrogram) that even more closely align frequency bins with what we perceive as human listeners which can be beneficial for tasks where perceptual relevance is critical [2, 3, 9, 45].

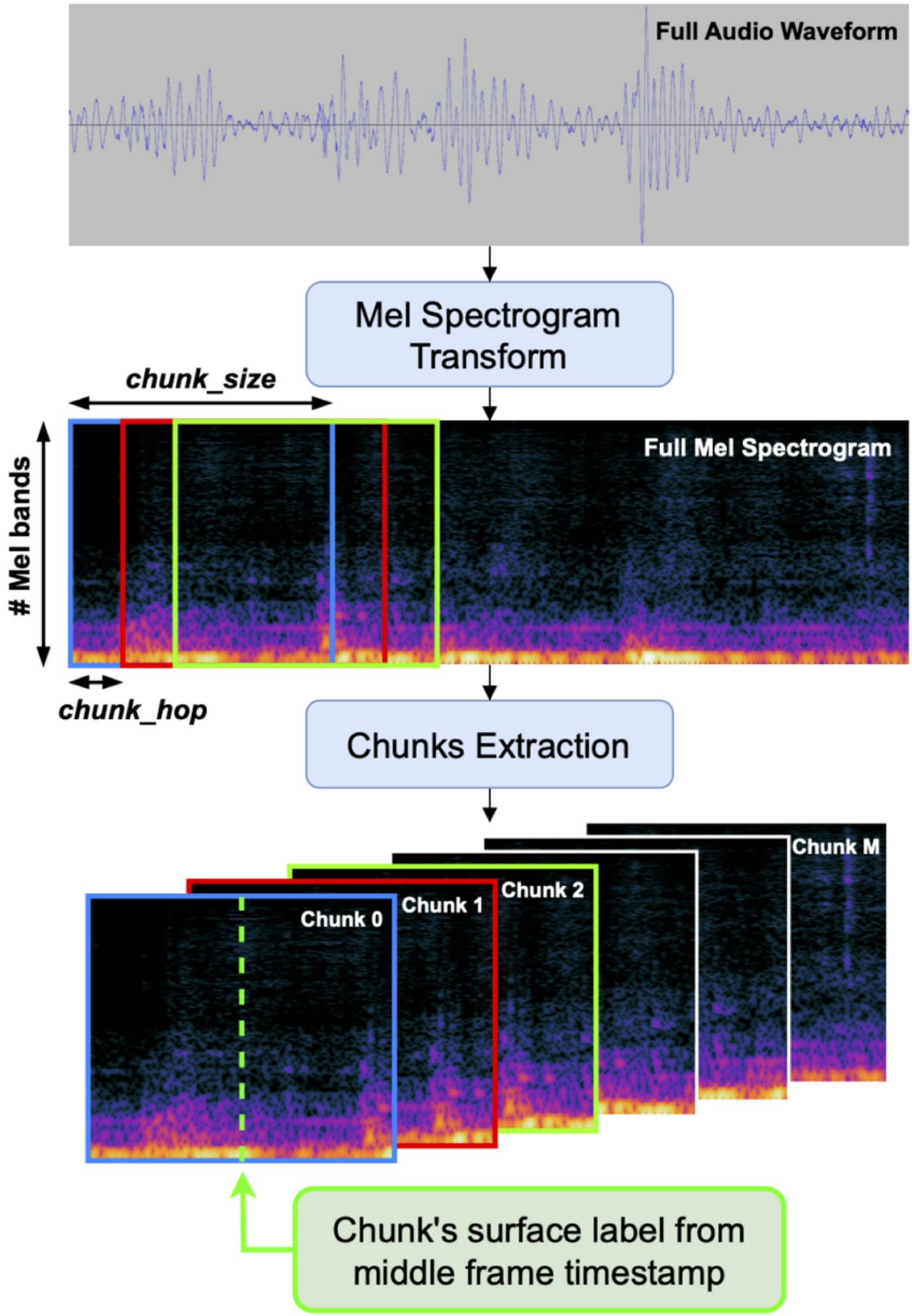


Fig. 9 Data preprocessing and Mel-spectrogram feature extraction pipeline: raw waveform → framing & windowing → FFT / spectral transform → Mel filterbank → log amplitude representation

Alongside the Short-Time Fourier Transform previously mentioned, the Constant-Q Transform or CQT, also has logarithmically-spaced frequency bins, allowing higher resolution for lower frequencies while having coarser resolution for higher frequencies. For this reason, the CQT is well-suited for analysis of musical signals that possess a strong harmonic structure [2, 3, 45]. Finally, wavelet transforms allow for the analysis of an audio signal hierarchically, picking up short-term transients up until spectral envelopes that last longer [2, 6, 45]. Audio signal representations can, in many cases, also be represented in image form, and thus, could be directly input into convolutional neural networks (CNNs), which bridge the gap between engineered features and the deep learning pipeline [19–21, 39].

$$X(k,m) = \sum_{n=0}^{N-1} x[n]\, w[n-m]\, e^{-j\frac{2\pi kn}{N}},$$

where X (k, m) is the localized frequency spectrum at frame m. STFT-based Spectrograms present harmonic structures and transients that are essential for classification. Differences of the STFT spelling, Mel-spectrograms apply Mel-scaled filter banks, and the Constant-Q Transform (CQT) achieves logarithmic resolution to capture harmonic richness in music. Wavelets provide extra flexibility by capturing fine-grained and global characteristics.

As shown in Fig. 10, the different time-frequency methods reveal different characteristics of the signal: the STFT distinguishes between harmonic and transient events, with a fixed time-frequency resolution, while a wavelet or scalogram representation uses variation in scale, allowing better representation of both short time bursts (transients) and long time spectral envelopes. These representations help in differentiating between speech, music, noise, and overlapping events in radio broadcasts, by building richer, multi-scale representations about speachural makes of acoustic structure.

3.4 Modern Embeddings and Learned Features

The constraints of hand-crafted features in dealing with noisy, overlapping, and varied radio audio led researchers to adopt more data-driven methods. Deep learning has provided researchers with the ability to extract features directly from raw waveforms or spectrograms to develop learned embeddings that exceed classical descriptors as provided for deeps speech detection [46, 47]. Furthermore, other examples are CNNs learn hierarchical local patterns in spectrograms [20–22, 39], while recurrent models like LSTMs and GRUs can also learn temporal dependencies over long sequences [13–15]. In more recent years, transformer-based models that employ self-attention mechanisms yield state-of-the-art results and perform well with capturing long-range dependencies in broadcast streams [16, 17, 26].

Another advancement in this paradigm has been the introduction of pre-trained embeddings. Models like Wav-2Vec 2.0 produce self-supervised representations that are learned from large unlabeled corpora [31–33]. The embeddings from these models can be fine-tuned on small labeled datasets for downstream tasks. YAMNet, trained on AudioSet, produces universal representations of audio that could be applied to classification tasks across multiple domains [17, 35]. Spleeter provides source-separated representations of speech and music which can greatly enhance performance in overlapping conditions [29]. Pyannote was optimized for speaker diarization tasks [30]. The introduction of these models provides less reliance on hand-crafted features while also working somewhat across domains. Overall, the less reliance on hand-crafted features and better generalization across domains are particularly important features when dealing with radio audio with the wide variety of content and conditions.

$$Attention(Q,K,V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V,$$

Here K, Q and V represent query, key, and value matrices. Pretrained models, including Wav2Vec 2.0, YAMNet, and Spleeter, generate universal embeddings from massive corpora, offering robustness across multiple audio domains.

In the spectrogram visualized in Fig. 11, the audio is segmented into patches which are then linearly mapped to

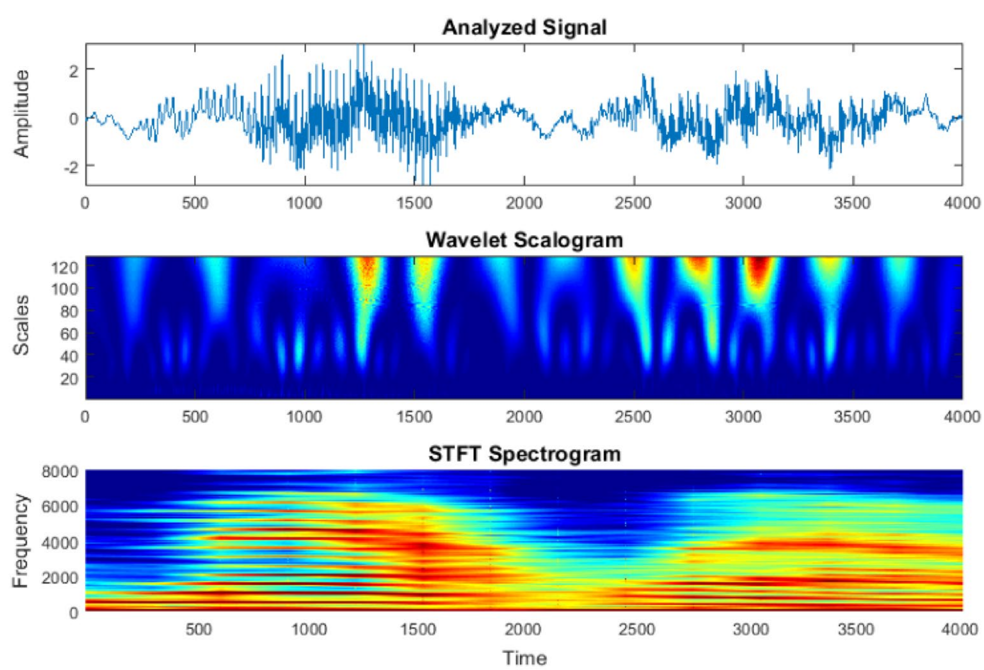


Fig. 10 Multiple time-frequency representation methods: top shows original waveform; middle shows STFT spectrogram; bottom shows wavelet / scalogram view

embedding vectors. The embeddings are then augmented with positional encoding to ensure the order is preserved before being passed through multiple transformer blocks. These blocks leverage self-attention in order to extract both short- and long-range dependencies through time and frequency. The resulting learned embeddings represent the audio signal effectively and can even be fine-tuned for specific tasks such as speaker diarization, speech vs. music discrimination, or event detection. This visual demonstrates the leap from engineered features to learned embeddings, while also validating that transformer architectures generalizes and transfers its features derived from real world radio audio data.

3.5 Dimensionality Reduction and Selection

The additional features added in increase the difficulty of high-dimensionality which can slow down computing time and may even lead to overfitting. High-dimensionality has usually been tackled with algorithms like Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) [2, 3, 45], however, there has been a shift towards learning a condensed latent representation with autoencoders and/or neural compression algorithms for efficiency in computing by retaining and preserving discriminability and/or invariance of the representation to maintain the advantage of rich feature development [18, 19, 48]. Selecting only the best-featured mechanics promotes efficiency of the model for obtaining accuracy with less complexity [18, 49].

3.6 Challenges in Feature Extraction for Radio Audio

Even with improvements, radio audio feature extraction still represents a tough problem. Overlapping acoustic events frequently smear temporal and/or spectral boundaries, making it challenging to disentangle sources using both handcrafted features and learned features [2, 18, 23, 29]. Broadcast contexts are noisy and unpredictable distortions can degrade both classical descriptors as well as the derived embeddings from trained networks [3, 35, 36]. Furthermore, imbalance in training datasets can present complications as rare classes like jingles or advertisements may not be included frequently enough in the training [27, 28, 41]. Lastly, related to the former issues are the requirements for real-time processing require computationally expensive transformations to be avoided, creating a trade-off between accuracy and process [31, 34, 38]. The above issues reinforce the need for the design of robust, scalable, and lightweight feature extraction facilities for the use of radio audio analysis [17–19].

4 Rule-Based Approaches

In the time before machine learning and deep learning were prevalent, the first attempts to audio segmentation and classification were largely aspiration and rule-based methods. These methods were based on deterministic heuristics, handcrafted features, and statistical thresholds that were easy to understand and computationally inexpensive [2, 5, 6, 45]. Although no longer advanced by current standards, rule-based methods introduced the core principles of energy tracking, spectral analysis, rupture detection, and speaker diarization, some of which remain relevant to hybrid pipelines today [4, 5, 11, 12]. Their usefulness should be considered not only in terms of historical importance, but also in terms of usability in low resources and in use as baseline methods for more advanced approaches [2, 4, 5]. Some of the rule-based approaches are tabulated in the form of Table 1

4.1 Time Series Analysis

An established way of working with rule-based segmentation is to apply a time series model, where an audio signal is

Fig. 11 Transformer-based audio embedding pipeline: spectrogram patches are linearly projected with positional encodings and processed through stacked transformer blocks to yield learned embeddings for downstream segmentation and classification tasks

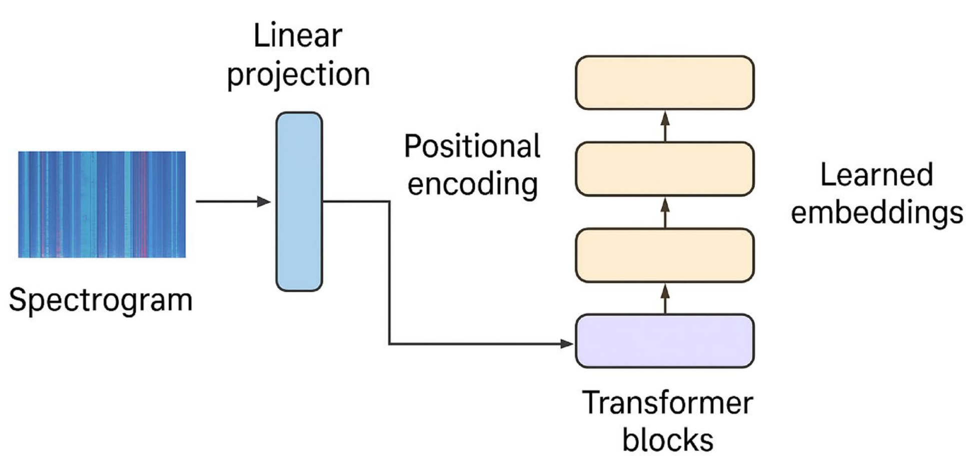


Table 1 Summary of core techniques for audio segmentation and classification

Technique Name	Brief Description / Principle	Key Application in Audio Segmentation / Classification
Time Series Analysis	Treating audio as sequential data evolving over time.	Detecting temporal patterns, trends, and changes; foundation for ML/DL models.
Speaker Diarization	Segmenting and clustering speech to identify "who spoke when."	Structuring multi-speaker recordings; enhancing transcription and recognition.
Intensity (dB) Detection	Measuring loudness levels and their variations.	Silence detection, loud event spotting, and presence thresholds.
MFCCs	Representing short-term power spectrum on a perceptual Mel scale.	Capturing timbre and phonetic traits for speech/music recognition.
Timbre Features	Characterizing perceptual sound quality beyond pitch/loudness.	Differentiating instruments, voices, and sound sources.
Chroma Features	Mapping audio into 12 pitch classes independent of octave.	Harmonic/melodic analysis, chord and tonal structure recognition.
Temporal Change Analysis	Monitoring evolution of audio features over time.	Segment boundary detection, rhythm tracking, dynamic shifts.
Power Spectral Density (PSD)	Measuring frequency power distribution.	Noise characterization; distinguishing signal vs. background.
Wavelet Transform	Multi-resolution decomposition of signals.	Balancing time-frequency detail; effective for speech/music transitions.
STFT	Local sinusoidal frequency-phase decomposition.	Generating spectrograms, foundational for DL feature pipelines.
Rupture Detection	Identifying abrupt signal discontinuities.	Structural segmentation (e.g., song sections, ads, pauses).

interpreted as a time series of statistical observations [2, 3, 45]. The purpose is to detect change points (the point (s) at which a distribution has changed) as shown in Fig 12. Short-time energy tracking and autocorrelation are commonly available approaches that are applied in a sliding window manner to the signal [2, 5]. In practice, change points that correspond to sudden changes in energy levels often signal transitions between silence, talking, music, or jingles as an example when you listen to the radio. More formally, the segmentation process entails comparing two probability distributions (or distributions of the measurements you have) before and after a candidate boundary point t (if a boundary point exists) [2, 5, 45].

$$P(x_{1:t}) \neq P(x_{t+1:T}),$$

The differences are measured using statistical tests like log-likelihood-ratios. Time series approaches are easy to understand and easy to implement, and they do not require training data, making them suitable for early real-time monitoring systems. Unfortunately, they rely on strict statistical assumptions and suffer from maximum degradation in noisy or overlapping acoustic conditions.

Time series-based method for audio segmentation, as revealed from Figure X, involves statistical-tracking of the audio signal via metrics such as short-time energy and autocorrelation. The figure shows how the raw audio waveform will overlay a profile of energy, and sudden drops/releases in energy would hint at possible boundaries between different acoustic events or signals. While an admittedly basic method, this visualizes the general ideas behind change point detection—clear shifts in amplitude characteristics would typically correlate to clear perceptual shifts, such as

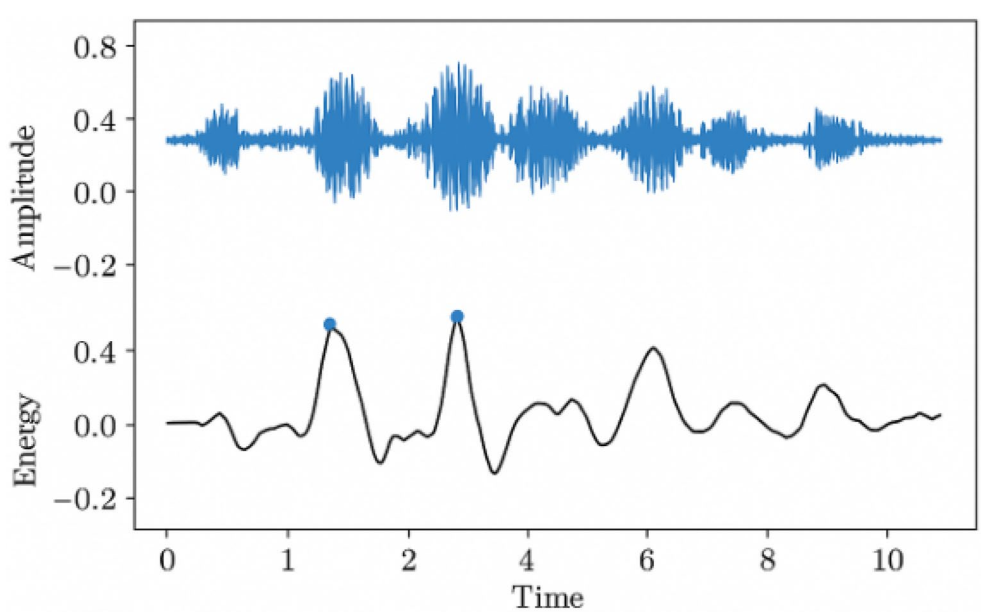


Fig. 12 Time series–based audio segmentation. The raw waveform is analyzed with short-time energy (STE), where change points (vertical markers) indicate transitions between silence, speech, and music segments

from silence to speech or music. All of these communicating processes (with visuals and numbers) helps situate how rule-based segmentation can be useful as a building block for more sophisticated and data-driven models in broadcast and noisy environments.

4.2 Speaker Diarization

Speaker diarization, essentially the task of figuring out 'who spoke when', originally revolved around systems that utilized rules [4]. These rule-based systems were systems that combined the idea of energy thresholds to identify segments of speech coupled with similarity metrics to cluster speech frames together that belonged to the same speaker [5]. With these systems, features (e.g., MFCCs) were modeled by Gaussian distributions, and measures such as Kullback–Leibler divergence were used to represent similarity with respect to difference or distance between speakers [45]. Change points were introduced based on criteria such as Bayesian Information Criterion (BIC) or Generalized Likelihood Ratio (GLR) which ask the question whether it is more likely to model two segments of speech separately in a mixture model than to model them jointly [4]. While such systems were interpretable, they were vulnerable to overlapping speech and the wide variability in speaking styles and conditions often found in broadcast radio [12]. Similarly, the presence of other audio events, such as background noise or music, had a degrading effect, creating a need to manually adjust thresholds based on context [11].

As shown in Fig. 13, the speaker diarization process starts with voice activity detection, which distinguishes segments of speech from silence or background noise. Features, typically derived from MFCCs or embeddings, are then modeled and compared using measures of similarity, such as Kullback–Leibler divergence. Candidate segment boundaries are determined through statistical assessments using the Bayesian Information Criterion (BIC) or Generalized Likelihood Ratio (GLR). In the last step, clustering joins similar frames to produce labels indicating 'who spoke when'. The efficiency of the modular approach allows for early diarization models to be interpretable, but exposes limitations related to overlapping speech, different acoustic environments, and noisy broadcasts.

4.3 Intensity-Based Detection

The simplest format of rule-based segmentation could be one that is based on signal intensity or energy level measured in decibels (dB). After calculating the short-time energy (STE) of each frame, segmentation systems would be able to detect silent pauses, abrupt sounds and high-energy events such as jingles and advertisements [2, 5, 45].

$$E_m = \sum_{n=0}^{N-1} |x_m[n]|^2$$

If E_m crosses a custom threshold, this indicates the start of a boundary. In practice, using various threshold levels, speech

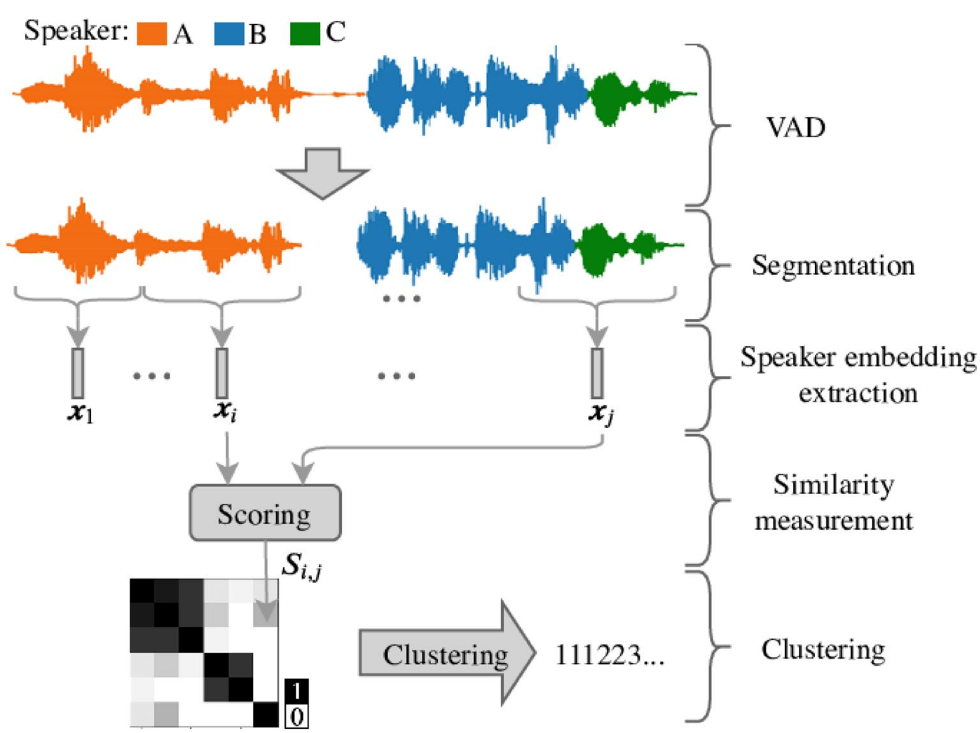


Fig. 13 A typical speaker diarization pipeline consisting of voice activity detection (VAD), feature extraction (e.g., MFCCs), clustering with similarity measures, and post-processing to assign "Who spoke when"

can be separated from silence or jingles can be identified, as jingles often have sharp intensity peaks [5]. However, intensity threshold limits are very sensitive to environmental conditions; the threshold that works in one broadcast or recording may be useless in another due to different playback volume normalization or background noise [11]. Because of these extra factors, the use of intensity thresholds can make them unreliable in large scale systems [12].

4.4 Spectral Features

As researchers sought to expand their analysis beyond simple intensity, they looked for spectral descriptors that examined audio by analyzing its frequency content. Mel-frequency cepstral coefficients (MFCCs) ultimately became the most popular, simulating human auditory perception by mapping linear frequencies to the perceptual Mel scale [2, 3, 6, 45, 50].

$$M(f) = 2595 \log_{10}\left(1 + \frac{f}{700}\right)$$

Chroma features, which project spectral energy into 12 pitch classes, were particularly useful for music detection since they capture harmonic constructions that set music apart from speech [9]. Timbre coefficients quantify the texture or "color" of sounds, and are used to distinguish between natural speech and artificial jingles or environmental effects [7, 8]. Rule-based systems normally compared these spectral features with either Euclidean distance, cosine similarity, or a sensitivity threshold that was manually tuned [2, 45]. Spectral features aided discrimination of classes, however, still exhibited sensitivity to noise and were not adaptable to new broadcast domains [3, 11].

Table 2 Strengths and limitations of rule-based methods for audio segmentation and classification

Method	Strengths	Limitations
Time Series Analysis	Simple, interpretable, real-time friendly; no training data needed	Sensitive to noise; weak discrimination for complex audio
Speaker Diarization (Rule-based)	Provided first diarization pipelines; interpretable clustering	Overlaps and noise reduce accuracy; threshold tuning required
Intensity (dB) Detection	Effective for silence detection, jingles, and abrupt high-energy events	Thresholds environment-dependent; fails in noisy or normalized streams
Spectral Features (MFCC, Chroma, Timbre)	Capture perceptual and tonal cues; useful for speech–music separation	Noise sensitive; manual feature design; limited generalization
Temporal Rupture Detection	More principled and statistical; captures subtle transitions	Computationally expensive; impractical for large-scale real-time monitoring

4.5 Temporal Change and Rupture Detection

An additional statistically acceptable family of rule-based methods is that of temporal rupture detection as it pertains to identifying sudden changes in the distribution of features over time. For example, sliding windows of MFCCs or energy features are compared across candidate boundaries using a likelihood-based method [2, 5, 45]. One common method is the Bayesian Information Criterion (BIC), which considers the model cost of modeling two adjacent segments separately versus modeling them jointly [4]. The BIC difference can be stated as:

$$\Delta BIC(\tau) = L(X_{1:\tau}) + L(X_{\tau+1:T}) - L(X_{1:T}) - \frac{\lambda}{2} k \log N,$$

where $L(\cdot)$ is log-likelihood, k the number of free parameters, and λ a penalty term controlling for model complexity [4, 12]. A positive ΔBIC indicates that there is likely a segmentation boundary at τ. Rupture detection methods offered greater stability compared to purely intensity or spectral thresholds but the computational burden presented challenges for broader applicability of real-time broadcast monitoring [5, 11].

4.6 Comparative Analysis of Rule-Based Methods

Rule-based systems provide the benefits of simple designs, interpretability, and low processing requirements that make them especially desirable for initial broadcast monitoring and embedded systems, which may have minimal computing resources [2, 5, 45]. These systems perform well for identifying "clean" transitions between programmed content, such as silence to speech, or music to advertising, and can even act as lightweight baselines, or effective sensor tools, in hybrid pipelines [4, 5]. However, their reliance on fixed thresholds, and handcrafted features results in impermanence in noisy and diverse radio contexts [11]. When applied to overlapping speech, background noise, and variation in programming norms between broadcasts, these systems are quickly compromised [12]. To make matters worse, rule-based systems do not adapt with changes to content or scale to modern datasets of broadcast data [3, 11]. These points are compared in Table 2 as well.

5 Classical Machine Learning Approaches

The shift from rule-based methods to machine learning signified a major change in audio segmentation and classification [2]. The introduction of classical machine learning methods brought statistical rigor, data-driven learning, and flexibility to unknown and diverse conditions [3, 51]. Although still based on handcrafted features such as MFCCs, chroma, wavelets, and spectral descriptors, classical machine learning methods replaced deterministic heuristics with probabilistic models, and/or discriminative classifiers improving generalization over segments and paving the way for deep learning architectures [18, 52]. Some of the most prominent classical machine learning methods were: Hidden Markov Models (HMMs), Gaussian Mixture Models (GMMs), Support Vector Machines (SVMs), and ensemble methods such as Random Forests, each with their benefits and limitations [45].

5.1 Hidden Markov Models (HMMs)

Hidden Markov Models introduced one of the first probabilistic frameworks for audio segmentation, especially in tasks that involve sequential modeling, such as speech recognition and broadcast analysis [45]. An HMM represents audio as a sequence of hidden states, each of which corresponds to a class like speech, music, or silence as developed in [53, 54]. Each hidden state emits observable features with certain probabilities. The joint probability of a sequence of states Q and a sequence of observations O is given by

$$P(Q, O) = \pi_{q_1} \prod_{t=2}^{T} a_{q_{t-1} q_t} \prod_{t=1}^{T} b_{q_t}(o_t),$$

where π_{q_1} denotes the initial probability of state q_1, a_{ij} are transition probabilities between states, and $b_i(o_t)$ defines the likelihood of observing feature o_t given state i [3, 45].

This probabilistic structure allowed HMMs to capture temporal dependencies in audio streams and improve upon static rule-based approaches [2]. In radio audio, HMMs were the basis for speech–music segmentation and speaker change detection efforts [4]. However, whilst they improved upon prior work, their reliance on Gaussian assumptions and handcrafted features were not robust to the complexities of broadcasts [11].

5.2 Gaussian Mixture Models (GMMs)

Gaussian Mixture Models augmented HMMs by establishing a relatively flexible model of the probability distribution of the acoustic features [27]. A GMM represents the density of feature space using a weighted sum of Gaussians, expressed as

$$p(x) = \sum_{i=1}^{M} w_i \mathcal{N}(x | \mu_i, \Sigma_i),$$

where w_i are mixture weights, μ_i are mean vectors, and Σ_i are covariance matrices [3].

GMMs were particularly efficient for speech–music discrimination and speaker verification problems in their ability to model multimodal distributions [4]. By using GMMs in conjunction with HMMs, researchers developed GMM–HMM hybrid systems that enable most initial speech recognition technologies to function properly [2]. GMMs had limitations as well, particularly in sensitivity to initialization and risk of overfitting when little training data was available [45]. In addition, scaling to high-dimensional feature spaces was computationally expensive, such as concatenated MFCCs or spectral flux [11].

In Fig. 14, we illustrate how a feature window (e.g. MFCC, spectral vectors) are scored under Gaussian mixture models which provide the emission likelihoods, and arranged via HMM transitions into likely class sequences. The decoding (e.g. Viterbi) determines the most likely sequence of hidden states (speech/music/silence) to describe the process. This is another way to say it is analogous to what was described in the Classical ML section, and shows how generative statistical methods replaced more simple heuristics in audio segmentation and labeled temporal and probabilistic structure.

5.3 Support Vector Machines (SVMs)

Support Vector Machines provided a discriminative alternative to generative models by directly optimizing boundaries between classes in feature space [23]. Given training data (x_i, y_i), an SVM learns a decision function

$$f(x) = \text{sign}(w \cdot x + b),$$

that maximizes the margin among classes. For non-linear problems, kernel functions can be utilized, such as the radial basis function (RBF):

$$K(x_n, x_i) = \exp\left(-\gamma \|x_n - x_i\|^2\right).$$

SVMs produced state-of-the-art performance on binary classification problems like speech–music separation, speaker vs non-speaker discrimination, and environmental sound recognition [21]. They were particularly robust in utilizing high-dimensional handcrafted features such as MFCCs and wavelets [40]. However, training time scaled poorly with the amount of data, and the need for kernel selection required a careful tuning process, which ultimately made them less scalable to very large archive broadcast methodologies [39].

Figure 15 shows a standard classical ML pipeline in audio classification - starting from preprocessed audio, common feature extraction like MFCCs, chroma, spectral flux, etc.; a set of classifiers (SVMs, decision trees, Random Forests)

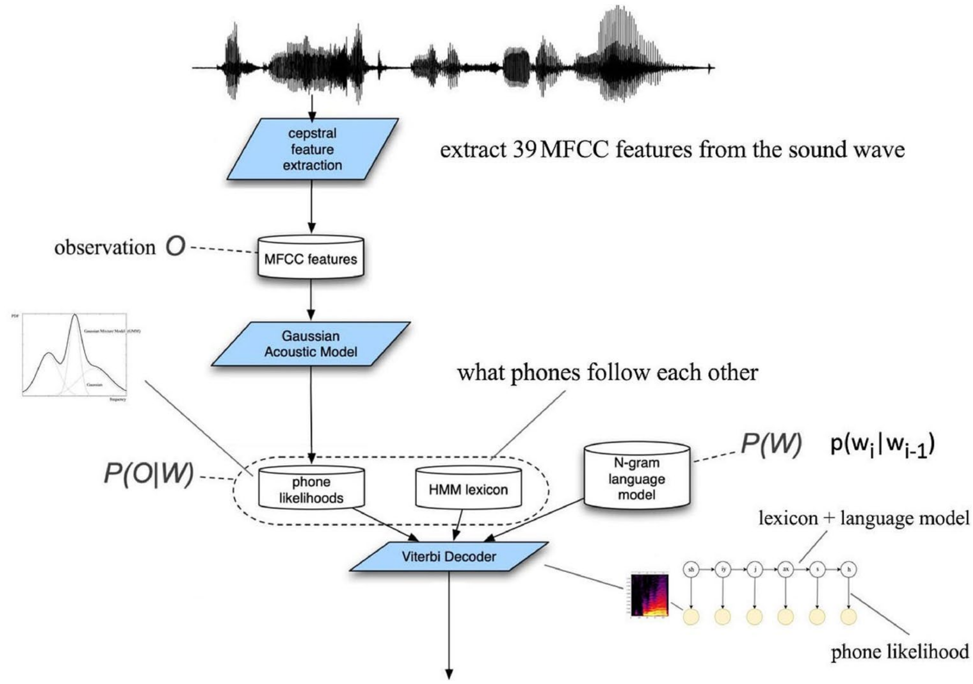


Fig. 14 High-level diagram linking HMM-GMM based speech recognition: feature windows → GMM emission scoring → HMM state transitions → decoded sequence outputs

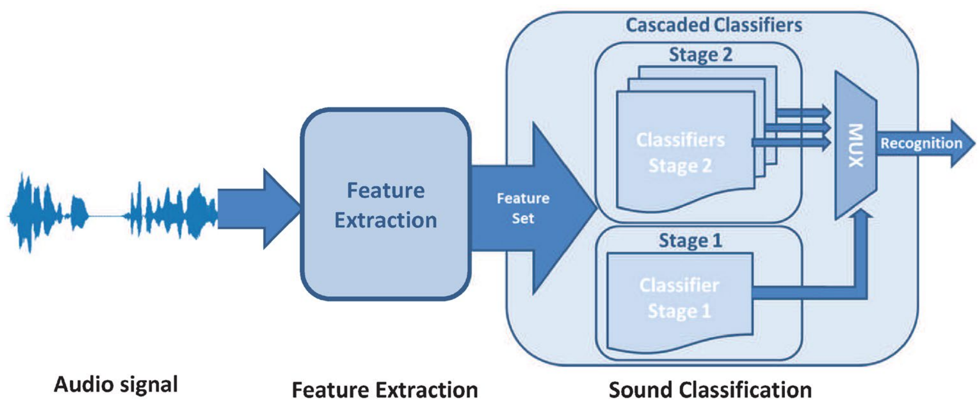


Fig. 15 Classical machine learning classification pipeline: preprocessing → extracting features → feeding into multiple classifiers like SVM, Random Forest, Decision Trees, and combining via ensemble methods

accept these features as input, and classification decisions may be combined or ensembled. This representation aligns with the section and its discussion of moving from handcrafted descriptor-based rule systems to statistical, discriminative and ensemble methods for segmentation and audio classification.

5.4 KNN, Decision Trees, and Random Forests

More straightforward classifiers, such as k-Nearest Neighbors (KNN), Decision Trees, and Random Forests, were equally important in early segmentation pipelines [4] as described in 3. KNN worked on a proximity-to-sample principle that assigned a class to the test samples based on the majority class of its nearest neighbors. It was simple enough to be a strong baseline classifier, but it was computationally inefficient for larger datasets [19, 55]. Decision Trees offered interpretable rule-based structures that recursively partitioned the feature space based on information gain, while being prone to overfitting and limited robustness to noise [17]. Random Forests attempted to address these issues by combining multiple Decision Trees trained on different subsets of the features and averaging the outcome of classifications for improved stability and generalization [34]. The ensemble methodology for Random Forests solved many of the concerns found in a Decision Tree classifier, especially when dealing with multi-class classification problems—as in the case for categorizing environmental sounds where classes were highly variable [22].

5.5 Comparative Analysis Across Datasets

Overall, the comparative performance of classical machine learning methods depends both on the dataset, and the task at hand. On the controlled datasets of both MUSAN and GTZAN, the GMM and HMM methods achieved accuracies

Table 3 Comparative analysis of classical machine learning algorithms for audio segmentation and classification

Algorithm	Strengths	Limitations	Typical Datasets
HMM	Models sequential dependencies; interpretable probabilistic framework	Relies on hand-crafted features; poor with overlaps	MUSAN, GTZAN
GMM	Captures multi-modal distributions; effective in speech–music separation	Sensitive to initialization; high-dimensional inefficiency	GTZAN, LibriSpeech
SVM	Strong discriminative power; robust with high-dimensional features	Expensive training; kernel selection critical	ESC-50, UrbanSound8K
KNN	Simple baseline; interpretable	Computationally costly at scale	ESC-10
Decision Trees	Fast, interpretable rules	Overfits; noise-sensitive	Small broadcast datasets
Random Forest	Improved robustness over trees; good generalization	Larger computational footprint	ESC-50, UrbanSound8K

of greater than 90% for speech–music discrimination [15]. On the uncontrolled environmental sound dataset: ESC-50 and UrbanSound8K, the discriminative methods of SVMs and some ensemble methods, such as Random Forests, outperformed the simple models [24]. In speaker diarization events, HMM, along with BIC based segmentation, was simultaneously more robust than rule-based systems alone, however the accuracy dropped drastically across noisy or overlapping speech conditions [5]. KNN, whilst still useful for smaller problems, was not appropriate for large-scale datasets like AudioSet, where computational cost became infeasible [12]. The results show that no individual method either performed the best, or was dominant across domains; ensembles and hybrid methods would provide the best accuracy vs computational efficiency trade-off.

In Fig. 16, we illustrate a process involving multiple-classifiers, where various classical models - like SVMs, Random Forest, or Decision Tree - are trained individually on the feature vectors, and their outputs are aggregated (voting, weights, or stacking). This compares the performance of single classifiers (SVM, KNN, etc.) and ensemble methods. Combining classical methods can demonstrate how performance can improve generalization and robustness across audio classes and in the presence of noise.

6 Data-Driven and Deep Learning Approaches

The emergence of deep learning has fundamentally changed the area of audio segmentation and classification. Previous use of genetic algorithm (GA) [56], learning-based approaches [57], Bayesian methods [58] and other transformations [50, 59] applied for audio and speech signals has evolved in many areas of studies. In contrast to rule-based or classical machine learning approaches that utilize handcrafted features, deep learning allows systems to learn representations of features hierarchically from raw waveforms or spectrograms. This data-driven approach provides significant improvements in robustness, generalization, and scalability, especially in hard cases encountered in broadcast contexts, which often involve overlapping speech, music, noise, and advertisements. Deep learning approaches can be categorized into convolutional and recurrent neural networks, transformer architectures, and transfer learning-based models that leverage large scale pretrained embeddings [38].

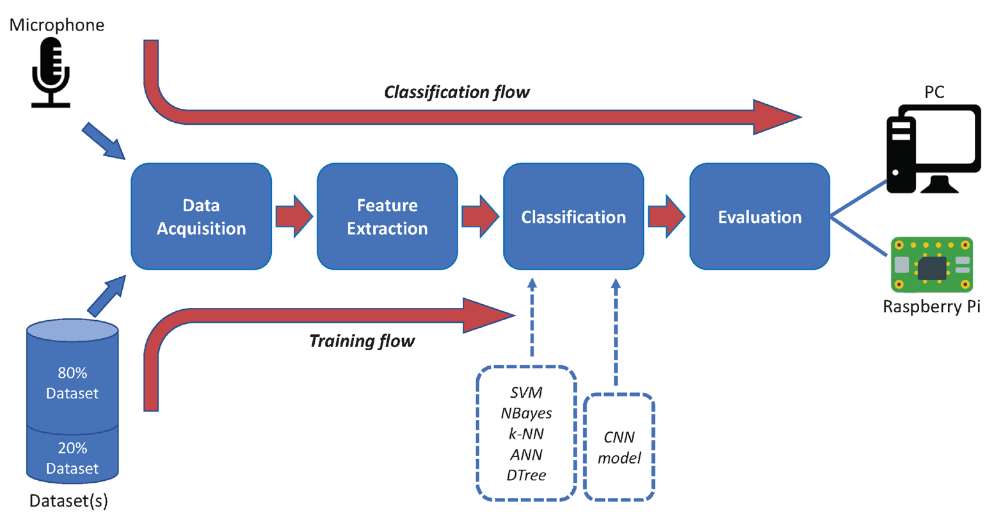


Fig. 16 Classical multi-classifier audio classification flow in the MosAIc framework illustrating multiple classifier types and ensemble decision

6.1 Convolutional and Recurrent Architectures

Convolutional Neural Networks (CNNs) have been instrumental for audio analysis because they can extract local hierarchical patterns from time–frequency representations (often called spectrograms) [38]. A 2D convolutional layer implements filters with a perceptive field across the entire spectrogram, where patterns with different effects (e.g. harmonic patterns, onset transients, and timbral textures) are extracted [60]. Formally, the convolution operation is shown as:

$$y_{ij} = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} x_{i+m,j+n} \cdot w_{m,n}$$

where x represents the input spectrogram, w denotes the filter weights, and y is the feature map. Deeper CNN architectures allow for increasingly abstract representations, which facilitate robust classification of a variety of audio classes.

Although CNNs are proficient at local feature extraction, they struggle in modelling long-range temporal dependencies. This shortcoming resulted in the application of RNNs and gated versions of them. Long Short-Term memory (LSTM) networks featured memory cells that maintained long-term context through the controlling information by way of input, forget, and output gates. An LSTM updates its hidden state at time step t as follows [1]:

$$h_t = o_t \odot \tanh(c_t), \quad c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

where i_t, f_t, o_t are input, forget, and output gates.

Gated Recurrent Units (GRUs) provide an easier option and reduced computation cost with roughly comparable performance. CNN–RNN hybrids merged local spectral feature extraction with temporal modeling [2]. and temporal modeling resulting in state-of-the-art performance criteria in discrimination of speech/music, detection of events, and classification of acoustic scenes.

The hybrid CNN–RNN model benefits from the different strengths of convolutional and recurrent networks, as illustrated in Fig. 17. First, convolutional layers analyze the input spectrogram and derive some discriminative local patterns (e.g., harmonic structures and timbral textures). These representations are then forwarded to recurrent layers (LSTM or GRU), which can encode the sequential aspects of the information and model the temporal dynamics of the audio over multiple frames. The dual-stage architecture enables the framework to model both localized spectral characteristics as well as global temporal dependence structure on the input, providing good performance on tasks such as speech–music discrimination, sound event detection, and acoustic scene classification.

6.2 Transformers for Audio

Transformers represent a substantial improvement by replacing recurrence with self-attention, addressing the long-range dependency modeling problem more effectively [1]. Transformers in audio processing think of spectrogram frames or waveform embeddings as a sequence of tokens and compute the pairwise interactions over the entire sequence. Attention mechanism is defined as:

$$Attention(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V,$$

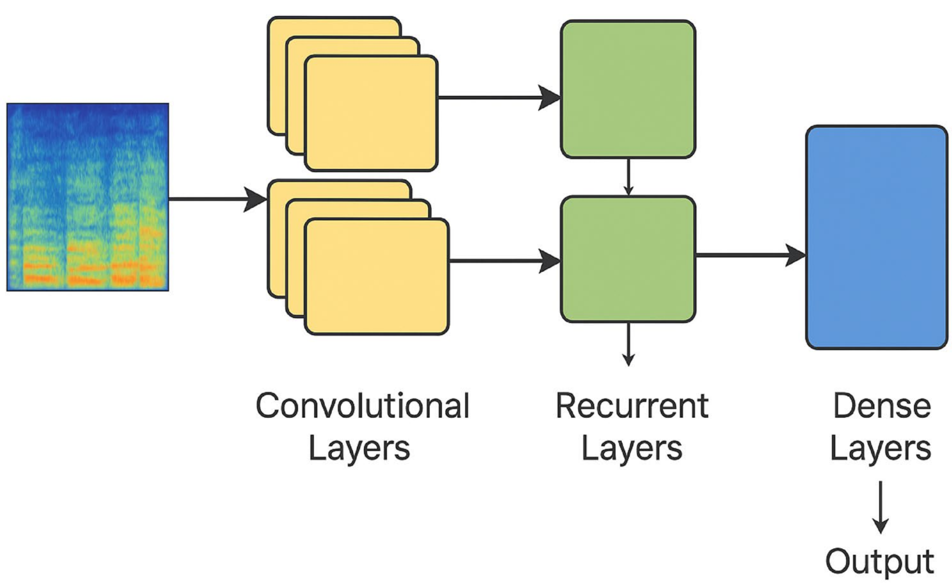


Fig. 17 Hybrid CNN–RNN architecture for audio analysis, where convolutional layers learn local spectral features and recurrent layers (LSTM/GRU) capture long-range temporal dependencies

where Q, K, and V represent query, key, and value matrices, and d_k is the dimensionality of the keys.

This function allows transformers to focus on relevant aspects of an auditory stream, providing a better means of capturing global structure from the data compared to a CNN or RNN. In this regard, architectures like AST (Audio Spectrogram Transformer) and HuBERT have greatly surpassed all models before it on tasks ranging from speech recognition to speaker diarization to sound event detection [35]. Additionally, transformers' ability to train efficiently on unprecedented amounts of data, while allowing for parallelizable calculations, have contributed to rapid increases in use and popularity over the last few years.

The spectrogram representation, shown in Fig. 18, is broken into small patches and projected into embeddings that are enhanced with positional encodings so that they retain the temporal order of the audio stream. The embeddings are then fed into the transformer encoder layers that encompass self-attention and are employed to model pairwise interactions over the entire sequence. The contextual embeddings produced from the transformers then provide local and global dependencies in the audio stream which solves the issue of CNNs and RNNs long-range modeling. This follows the description of how transformer architectures, the AST and HuBERT change the landscape of audio processing in the previous section, and it allows for models to dynamically emphasize relevant parts of the sequence and surpass human performance on tasks like speaker diarization and detection of sound events.

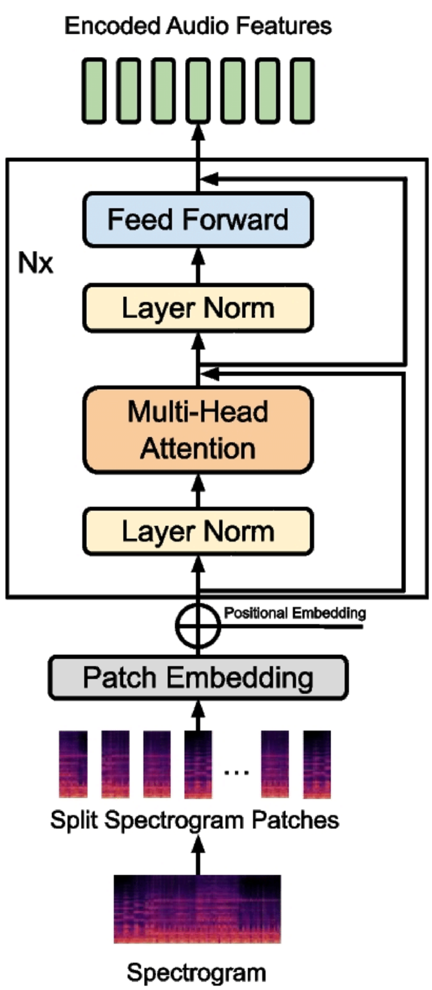


Fig. 18 Transformer-based audio encoder: spectrogram patches are linearly projected to embeddings, positional encodings are added, and transformer encoder layers compute context-aware features

6.3 Transfer Learning and Pre-Trained Models

In the field of audio analysis, transfer learning is valuable, particularly due to the infrequency of domain-specific labeled datasets. Transfer learning provides the ability to take advantage of large, pretrained models, wherein this transfer enables researchers to learn embeddings of audio data that are applicable across different tasks.

- The model YAMNet, which is trained on AudioSet, produces embeddings of sound sources using a dataset containing over 2 million audio clips that are mapped to 527 sound classes. The model outputs embeddings of sound sources, following learning, that show similarities of embeddings and indeed verify audio classification to power-of-10 classification accuracy [2].
- Wav2Vec 2.0 produces contextualized speech representations via self-supervised learning on raw waveforms. This method enables the fine-tuning dimension for Wav-2Vec 2.0 to power successfully on speech recognition and speaker diarization [1].
- CLAP, or Contrastive Language-Audio Pretraining, is also an alignment of audio signals with natural language. This alignment borrows from the literature known as zero-shot classification, using text queries to classify sound sources. Thus, multimodal learning allows the applicability of deep audio models to multiple audio-related tasks, such as captioning and retrieving audio [16].

Collectively, the language features in these pre-trained embeddings reduce the reliance on large labeled datasets and expedited the development of a domain-specific purpose for analyzing broadcast and audio data.

6.4 Specialized Tools for Audio Segmentation and Classification

A number of specialized deep learning frameworks have been developed, some of which are already being widely used in practice. The first is YAMNet, which uses Mobile

Net as the basis for lightweight embeddings that are general purpose for classification tasks [38]. Next is Spleeter which was developed by Deezer, a music related app. Spleeter allows for the separation of music sources, so that vocals, drums, and instruments can be isolated and confused less when performing downstream classification, for example [1]. Pyannote can be used for speaker diarization, where it offers a collection of neural embeddings and a clustering method to segment the speech data, and it has shown good results with speaker segmentation [2]. The last of these is Wav2Vec 2.0 methods, which have had dramatic impacts in speech processing for offering self-supervised representation learning from raw audio, and it is possible to completely remove engineered features [36]. In summary, these examples highlight the benefits of using specialized, pretrained architectures when addressing a number of broadcast audio tasks.

As demonstrated in Fig. 19, the Pyannote framework distinguishes the embedding extraction process from the segmentation process: first, audio input is processed to generate speaker embeddings, and then those embeddings are passed on to the segmentation module (in time) to apply speaker labels. This facilitated distancing is consistent with how many specialized tools separate representation and decision processes to achieve modularity and robustness—completely consistent with the brief description you give of Pyannote (and also YAMNet, Spleeter, and Wav2Vec 2.0)

6.5 Comparative Analysis of Deep Learning Approaches

Numerous datasets have been used to evaluate deep learning methods, which consistently outperformed classical ML methods in those evaluations. For example, YAMNet and CLAP have reported high accuracy on Audio Set's hundreds of sound classes using pretrained model techniques [1, 38]. Wav2Vec 2.0 also improved substantially on generic GMM-HMM baselines on LibriSpeech in the speech recognition context [2]. With speaker diarization, Pyannote embeddings with clustering methods improved upon much more classic BIC-HMM methods while achieving a still strong diarization error rate [36]. With environmental sounds, CNN and transformer architectures performed better than SVM and Random Forest baseline methods on both ESC-50 and UrbanSound8K studies while often exceeding a 90% accuracy rate [13, 16]. The described methods are also compared in the form of Table 4.

7 YOHO Algorithm (You Only Hear Once)

Most approaches in deep learning for audio segmentation use a frame-based approach, where models provide the label for short overlapping frames of audio that are typically 20–40 milliseconds long [1]. While these methods can efficiently learn local spectral and temporal patterns, the redundancy and inefficiency inherent in the frame-based system limits overall accuracy. For examples, while each frame is independently classified, the post-processed predicted labels need to be aggregated for the model to determine fixed segmentation boundaries. Using aggregation increases the amount of computation required for the task, creates issues for real-time applications, and reduces the precision of the resulting segmented events. To resolve this difficulty, the You Only Hear Once (YOHO) algorithm has recently been proposed as a regression-based way to solve the segmentation problem, in which it reframes the problem of segmentation to simply predicting boundaries [38]. Instead of predictions at the frame level, YOHO predicts segment onsets and offsets all at once, representing a more fundamental shift in thinking about segmentation of audio.

The uniqueness of YOHO is based on how it is designed to work in an analogous fashion to the You Only Look Once (YOLO) object detection framework from computer vision [39] – a framework that, like YOHO, addresses the problem of detection in a single, complex step based on frames and

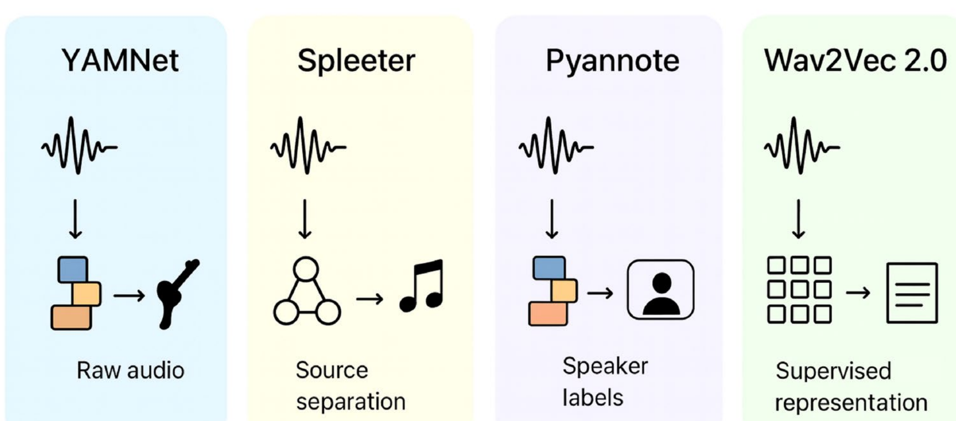


Fig. 19 Pyannote segmentation structure illustrating embedding generation and downstream segmentation modules

Table 4 Comparative analysis of deep learning approaches for audio segmentation and classification

Approach	Key Features	Datasets	Performance Highlights
CNN / CNN–RNN hybrids	Local spectral + temporal modeling	ESC-50, UrbanSound8K	>90% accuracy on ESC-50
LSTM / GRU	Long-term temporal dependencies	MUSAN, LibriSpeech	Improved diarization vs HMM
Transformers (AST, HuBERT)	Self-attention, global context	AudioSet, LibriSpeech	State-of-the-art speech recognition
YAMNet	Pretrained on AudioSet (MobileNet backbone)	AudioSet	General-purpose embeddings
Wav2Vec 2.0	Self-supervised waveform embeddings	LibriSpeech, VoxCeleb	Superior to GMM–HMM, SVM
Pyannote	Diarization-specific embeddings	AMI, VoxConverse	Low diarization error rate
Spleeter	Source separation for cleaner features	MUSDB18	Improves music detection
CLAP	Multimodal audio–text embeddings	AudioSet, LAION-Audio	Zero-shot audio classification

class probabilities and has been an important breakthrough in the object detection task. YOLO streamlined the object detection task by replacing complex object detection pipelines based on region eliminating proposals with a fast, single shot regression model which can predict the bounding boxes, and class probabilities, in real-time at the time of prediction. YOHO does the same for audio – whether able to describe spectrograms directly, or learned embeddings of audio, we take the segments of audio and slice them into temporal "grids," and for every grid the network regresses directly the onset and offset of the acoustic events with the respective class probability. So the need for frame-wise predicting and then merging the prediction is no longer need as a post-processing strategy, which drastically reduces the post-processing complexity.

In terms of architectural characteristics, YOHO uses a lightweight convolutional backbone, typically built on MobileNet [40], because it's a well-balanced trade-off between accuracy and efficiency. MobileNet uses depth wise separable convolutions, which reduce the computational cost while simultaneously maintaining substantial representational power. YOHO has a two-module architecture: the feature extractor (MobileNet backbone) and the detection head functional module. The feature extractor takes in spectrogram patches and uses them to create hierarchical feature maps.

The detection head outputs, for each temporal region, a set of parameters $(t_{on}, t_{off}, c_1, c_2, \ldots, c_k)$, where t_{on} and t_{off} denote predicted onset and offset times, and c_i are the probabilities for k acoustic event classes [38].

The regression of boundaries is optimized using the combination of mean squared error (MSE) loss which is supervised in temporal predictions and cross-entropy loss which is supervised in class probabilities [39, 40].

$$L = \lambda_1 \sum (t_{on} - \hat{t}_{on})^2 + \lambda_2 \sum (t_{off} - \hat{t}_{off})^2 + \lambda_3 \sum CE(c_i, \hat{c}_i),$$

where $\lambda_1, \lambda_2, \lambda_3$ are weights balancing the components.

YOHO is trained on annotated datasets that have exact segment boundaries. Exemplary datasets, like MUSAN, UrbanSound8K, and subsets of AudioSet, are used to test YOHO's generalization across speech, music, and environmental sounds [27, 28]. YOHO is unique in that it generates event boundaries directly, eliminating the processing of smoothing, thresholding, or median filtering, which is required in frame-based systems [1]. However, YOHO does utilize a small amount of non-maximum suppression (NMS) at the end of the regression, in order to suppress any duplicated event types, similar to that used in object detection [39].

As depicted in Fig. 20, YOHO converts the traditional frame-based parsing workflow into a direct regression alternative. The method begins with taking spectrogram patches into a lightweight feature extractor built on a framework like MobileNet, which yields hierarchical representations of the audio. The feature representations are then sent to a regression head, which concurrently predicts the onsets and offsets of events while also predicting class probabilities. Outputs are refined through non-maximum suppression (NMS) to remove duplicate predictions and obtain accurate segmentations. This design demonstrates YOHO's key contribution of eliminating frame-based classification and heavy post-processing for decluttering event predictions, which yields better temporal accuracy and allows real-time processing capabilities in broadcast monitoring and audio analytics.

Evaluations of YOHO's performance indicate that it outperforms traditional deep learning (DL) performance with frame-based models in accuracy while eliminating latency [1, 39]. For instance, YOHO measures higher F1 scores on UrbanSound8K while being faster. Similarly, YOHO provides more reliable and precise segmentation of speech (against); music (against) noise, with the MUSAN dataset [27, 28]. Anticipating studies, YOHO's strength is realized with radio broadcasts through identifying advertisements, jingles, and speakers in an extremely noisy setting [49]. Furthermore, the benefit of YOHO is in temporal precision; for instance, for frame-based systems - boundaries could be off by tens of milliseconds despite video relying on matching with the ground truth. In contrast, YOHO generates

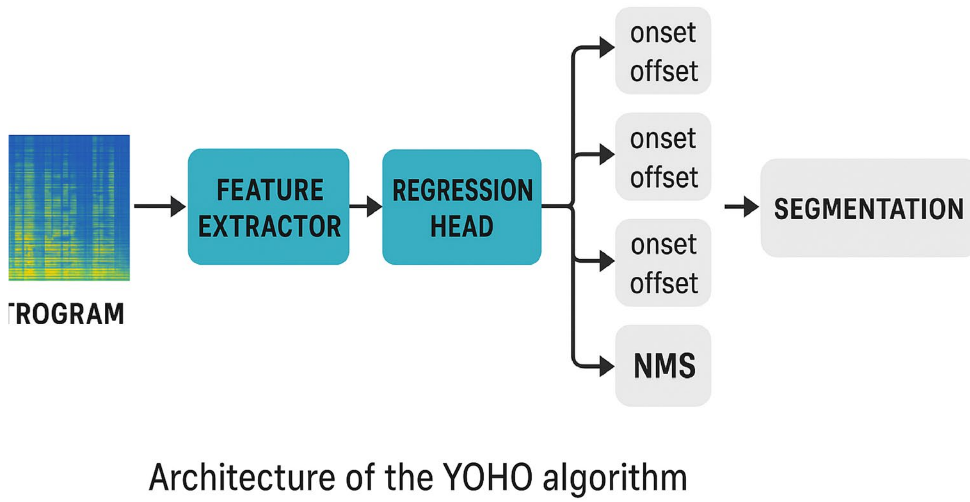


Fig. 20 Architecture of the YOHO (You only Hear Once) algorithm for audio segmentation, showing its MobileNet-based feature extractor, regression head for onset–offset prediction, and non-maximum suppression for final event segmentation

segmentation tighter (more closely) to the video (ground truth) [1].

One of the primary strengths of YOHO is its computational efficiency, which is critical for live broadcast monitoring. YOHO avoids not only the frame-level redundancy and computational costs associated with the model's post-processing, but also reduces its memory footprint and time-to-inference to make segmentation accessible in nearly real-time on multi-hour streams of audio on modern GPUs, and lightweight variants for embedded devices on MobileNet architectures [18, 39]. These computational efficiencies provide greater scalability and usability for applications where speed is as important as accuracy, especially in live radio, advertisement monitoring, compliance verification, and large-scale broadcast indexing applications [25].

In summary, YOHO's contributions reflect a relative shift from frame-based classification to boundary regression for audio segmentation. Based on the YOLO methods used in computer vision, YOHO is likely to provide more efficient prediction of onset-offset on a MobileNet backbone, maintaining better temporal accuracy, less post-processing, less time-to-inference, and ultimately more real-time capability [39]. Integrating the best aspects from vision-type segmentation and audio segmentation, YOHO is not only a pathway to a new trajectory in the state of the art, but also a domain for building hybrid models with boundary regression and transformer-based embeddings in multi-models, as the field progresses [1, 5].

8 Comparative Analysis and Discussion

The development of audio segmentation and classification methods signals a gradual shift from heuristics based on human-led crafting, to data-driven methods that learn complex feature hierarchies [2, 23]. What began, with clearly interpretable heuristics at the start and broad statistical modeling in classical machine learning models, transitioned into deep learning techniques that offer new sources of robustness and generalization for classification and segmentation [1, 13, 19]. In thinking about the latest development in this progression of methodologies, it is useful to consider the YOHO algorithm (discussed later), which allows segmentation to be framed as boundary regression [38]. The current section presents a comparison of these methods, through tabular summaries and discussion of relative strengths and weaknesses, across these educational models in their use for segmentation. To summarize, all analysis has been reported in the form of Table 5, Table 6, and Table 7.

8.1 Rule-Based vs. Machine Learning vs. Deep Learning vs. YOHO

Rule-based systems, like silence detection, or detecting energy levels, are interpretable and have less computational load, but they are sensitive to noise and heterogeneity of the signal [5, 36]. Classic machine learning models, in particular HMMs and GMMs, improved robustness by utilizing probabilistic modeling, but they still depended on handcrafted features [20, 23]. Deep learning, including CNNs, RNNs, and transformers, greatly outperformed these approaches by learning discriminative representations of raw audio, or spectrograms, directly from the signal [13, 19, 39, 40]. YOHO then improved these models by removing frame-level redundancy and regressing segment boundaries directly, thus combining accuracy and efficiency 1, 38].

This juxtaposition underscores the incremental trade-offs spanning the paradigms, which explains the effectiveness of hybrid architectures – combining rule-based preprocessing, deep models, and YOHO-style regression – for radio analysis.

Table 5 Comparative analysis of rule-based, classical ML, deep learning, and YOHO approaches for audio segmentation and classification

Approach	Strengths	Weaknesses	Typical Applications
Rule-Based	Simple, interpretable, fast	Poor generalization; fails in noisy/overlapping audio	Silence detection, coarse segmentation
Classical ML	Probabilistic/statistical modeling; interpretable	Relies on handcrafted features; limited scalability	Speech/music discrimination, diarization (small datasets)
Deep Learning	Learns hierarchical features; robust to noise; strong generalization	High computational cost; requires large datasets	Broadcast analysis, acoustic event detection, speech recognition
YOHO	Direct boundary regression; real-time applicability; reduced post-processing	Relatively new; limited large-scale benchmarking	Precise segmentation in real-time broadcast monitoring

8.2 Feature Extraction Methods Vs. Performance

Feature extraction is at the heart of any segmentation pipeline. Traditional hand-designed features, such as MFCCs, chroma, and ZCRs, are lightweight and interpretable but are not always robust in diverse conditions. Deep features from CNNs, transformers, or pretrained embeddings (for example, Wav2Vec 2.0, YAMNet) capture semantic richness and temporal dependencies.

The comparison illustrates that handcrafted features continue to have relevance for low-impact use cases, but that in all broadcast-scale use cases, deep embeddings outperform handcrafted features.

8.3 YOHO vs. Other State-of-the-Art Models

YOHO stands apart from classical frame-based deep learning models in its use of a boundary regression framework. While CNN–RNN hybrid systems and transformer-based classifiers predict frame-level labels and use smoothing, YOHO predicts the onset–offset times as well as class probabilities. This yields greater temporal precision at lower computational cost.

8.4 Recent Advances in Audio Segmentation and Classification

The field has witnessed significant progress in recent years, with advances spanning feature representations, neural architectures, and benchmark datasets. We have gone through some recent and most impactful contributions in the field, which are described below.

8.4.1 Feature Extraction Methods

Recent work continues to leverage Mel-frequency representations while exploring learned alternatives. MFCC features remain effective for environmental sound classification [61], while Mel spectrograms enable direct transfer from vision models [62]. A notable contribution by [63] has proposed time-frequency-wavelet fusion networks capturing multi-resolution acoustic characteristics. In recent studies, self-supervised approaches have matured considerably, which includes Wav2vec-AD [64] and Codec2Vec [65]. Wav2vec-AD integrates acoustic unit discovery with contrastive learning, while Codec2Vec introduces discrete token-based representations for efficient speech modeling.

Table 6 Comparative analysis of feature extraction methods for audio segmentation and classification

Feature Type	Examples	Strengths	Limitations	Performance Impact
Temporal Features	Zero-Crossing Rate (ZCR), Short-Time Energy	Simple, efficient	Sensitive to noise; limited discrimination	Good for speech/silence detection
Spectral Features	Spectral Centroid, Flux, Bandwidth	Capture frequency content	Require careful tuning; weak under overlap	Useful for coarse classification
Cepstral Features	MFCCs, LPC	Perceptually motivated; widely used	Discard fine temporal detail	Strong baseline for speech/music separation
Time–Frequency Features	STFT, Wavelets, CQT	Rich local context	Computationally expensive	Improves classification accuracy
Learned Embeddings	CNN filters, Wav2Vec, CLAP, YAMNet	Robust, scalable, generalizable	High data/training cost	State-of-the-art performance (> 90% ESC-50, >95% LibriSpeech)

Table 7 Comparison of deep learning architectures and YOHO for audio segmentation

Model	Segmentation Strategy	Strengths	Limitations	Performance Notes
CNN–RNN Hybrids	Frame-wise classification + post-processing	Strong temporal + spectral modeling	Redundant computations; post-processing needed	High accuracy but slower inference
Transformers (AST, HuBERT)	Token-wise classification with self-attention	Global context modeling; scalable	Heavy computation; large datasets required	State-of-the-art recognition
YOHO	Boundary regression (onset–offset)	Real-time capable; precise segmentation	Limited adoption; requires annotated data	Comparable accuracy with lower latency

Table 8 Feature extraction and self-supervised learning approaches

Method	Year	Type	Contribution
MFCC Features [61]	2024	Handcrafted	24 coefficients with deltas
TF-Wavelet Fusion [63]	2025	Hybrid	Multi-resolution analysis
Wav2vec-AD [64]	2024	Self-supervised	Acoustic unit discovery
Codec2Vec [65]	2024	Self-supervised	Discrete token representations

Table 9 Comparison of recent audio segmentation and diarization models

Model	Year	Architecture	Task	Key Innovation
YOHO [38]	2022	MobileNet CNN	Audio Segmentation	YOLO-style grid predictions
AST [66]	2021	Transformer	Audio Classification	Convolution-free attention
Audio Mamba [67]	2024	Mamba SSM	Audio Tagging	Linear complexity
LS-EEND [68]	2024	Transformer	Speaker Diarization	Streaming processing
Mamba-Diarization [69]	2024	Mamba SSM	Speaker Diarization	Efficient long-form
EEND-NAA [70]	2024	Transformer	Speaker Diarization	Non-autoregressive attractors
Enhanced DL [71]	2025	Hybrid NN	Speaker Segmentation	Multi-stage integration

Table 10 Benchmark datasets for audio analysis

Dataset	Year	Domain	Primary Use
DCASE 2025 S5 [73]	2025	Spatial Audio (2,290 files)	Spatial segmentation
DCASE 2024 Task 4 [74]	2024	Domestic Sounds	Heterogeneous SED
AudioSet [75]	Ongoing	General (∼2 M clips)	Pre-training/Tagging

The comparison encompassing these advances are listed in Table 8.

8.4.2 Deep Learning Architectures

Within these years, transformer-based models have achieved state-of-the-art results across diverse tasks. The Audio Spectrogram Transformer (AST) [66] established convolution-free audio classification with 0.485 mAP on AudioSet. Similarly, Audio Mamba [67] offers an efficient alternative with linear complexity for long sequences.

For speaker diarization, end-to-end neural approaches dominate. Example of these models is LS-EEND [68], which enables streaming diarization with online attractor extraction. Similarly, mamba-based segmentation [69] applies state space models for efficient processing. In another work, EEND-NAA [70] models introduces non-autoregressive attractors for parallel inference. An enhanced framework by [71] improves speaker segmentation accuracy through multi-stage integration.

Throughout these techniques, YOHO [38] remains influential, adapting MobileNet for YOLO-style audio segmentation with grid-based predictions directly outputting segment boundaries. The advances in the area are shown in Table 9.

8.4.3 Benchmark Datasets and Challenges

Dataset benchmarking is also one of the most crucial tasks to design efficient systems for audio detection and segementation. DCASE 2025 is one of the datasets, which introduces Spatial Semantic Segmentation of Sound Scenes (S5) [72], targeting joint detection and separation from multi-channel inputs. The S5 dataset contains 2,290 pre-mixed files [73]. Before this dataset, DCASE 2024 Task 4 [74] advanced sound event detection with heterogeneous supervision. AudioSet-Tools [75] provides taxonomy-aware dataset curation. The benchmark dataset comparison is shown in Table 10

Research studies have shown that YOHO can segmentation tasks at a level similar to or better than existing segmentation tools, and it does so in an efficient manner with respect to inference time, making it a more desirable option for real time broadcast monitoring and compliance verification.

9 Key Takeaways

In this work, we have found key findings from our cross-comparative analysis for audio detection and automatic time stamp recognition. One of the key findings is the efficiency analysis by transition from rule-based methods to state of the arts transformers based methods. Therefore, we have reviewed previous methods and analysed them on the basis of their strength and weaknesses:

- First, we found that rule-based methods can still provide value as preprocessing steps towards silence identification and coarse segmentation but cannot be considered an autonomous method capable of modern radio analysis [2].

- Second, the application of classical ML methods for our task are computationally efficient and explainable, but they do not have the flexibility or scale needed for broadcasting scale analysis for radio.
- After it, the results obtained from deep learning methods are found effective, yet they come with a heavier computational load and may not be feasible in less resourced deployment contexts.
- Finally, YOHO can act as a bridge between the power of deep learning calculations with computational efficiency, however, more benchmarking will be needed to assess the performance of YOHO on highly diverse datasets since it is new.

The key takeaway from our work is that we can use hybrid methods for our analysis. The hybrid-methodology and multi-method research and analysis design is likely the most effective way forward [26]. The hybrid-methodology employed rule-based preprocessing steps, ML classifiers for simple discrimination, deep learning for longer even toxic events, and YOHO for boundary regressions. This layered integration was likely successful because it provided a sufficient balance of accuracy, computational efficiency, and applicability in the "real-world", for the original goals and certainly addressed the primary issues Sect. 1 identified from the literature.

10 Challenges and Future Directions

Although advances have been made in audio segmentation and classification, the performance of existing methods continues to be hindered by a number of challenges in functioning with audio signals in real-world radio analysis. These challenges are not solely driven by the complexity of working with audio signals, but also driven by the computational and pragmatic limitations of deploying AI systems within live broadcast contexts. Researching towards addressing the important limitations of existing work requires attempting new and innovative research trajectories towards accuracy, generalization and efficiency.

10.1 Noise Robustness and Overlapping Events

Radio streams are very heterogeneous and may be corrupted by background noise, overlapping speech, mixtures of music and ads, and so on. When deep learning models that are trained on curated data can exploit a clean and controlled context, their robustness declines sharply in a noisy and uncurated ambient context. The class of overlapping acoustic events adds further complications as many models assume exclusive labels for each frame or segment. Future research needs to balance multi-label segmentation with classification enabling source separation model integration. Denoising models, such as deep generative models, or self-attention-based deep learning may also make classifying noisy input more robust for a single or multi-label setting. Novel combinations of heuristically derived boundary regression based on YOHO, and source separation (e.g., Spleeter) models may offer another promising way to address the ambiguity of overlapping signals within the broadcast monitoring context.

10.2 Data Imbalance and Annotation Cost

Large and accurately annotated datasets are critical for supervised deep learning models, but broadcast-domain-specific datasets that have precise segment boundaries are still quite limited. Annotation of audio streams can be not only time- and labor-intensive, but also quite costly when segment-level precision is required. A more pronounced imbalance is the result, where more common classes (e.g., speech, music) are overrepresented and rare events (e.g., jingles, political ads) are underrepresented. Future directions involve the development of weakly supervised learning. Rather than relying on frame-level labels, models can learn from clip-level labels. There are also self-supervised approaches, like Wav2Vec and HuBERT, that reduce reliance on manually annotated datasets. Semi-supervised active learning pipelines, where models request annotators only for the most informative samples, also have the potential to address the annotation bottleneck.

10.3 Generalization Across Domains

Models that are built on specific datasets typically do not generalize across broadcast domains, languages, or recording conditions. For example, a system trained on English radio, may not do very well when faced with multilingual broadcasts or broadcasts from regional channels (e.g. BBC, NPR) with further differences in vocal timbre and background noise. This is why domain adaptation and transfer learning is important. CLAP and Wav2Vec 2.0 are among the pretrained embeddings that show promise for cross-domain generalization, but these will need to continue to include unsupervised domain adaptation, adversarial training, and perhaps most importantly, will need to avoid the reliance on annotations and instead use meta-learning approaches that enable the models to adapt quickly with minimal fine-tuning. Globally deploying these models involves many uncertainties in terms of robustness across geography, language, and broadcast style, and this may be the most important question globally.

10.4 Real-Time Processing Constraints

Although accuracy is an essential requirement, practical applications, such as compliance monitoring and live broadcast indexing, require low latency processing. Frame-based deep learning models are typically very accurate but usually result in a noticeable delay in inference processing due to redundant computation and post-processing cost. The YOHO boundary regression solution partially resolves this issue, however researching lightweight architectures, model compression (quantization, pruning, distillation), and efficient inference engines that can be run on edge devices warrants further investigation. Real-time deployment also requires parallelization strategies in addition to distributed systems which effectively scale while consuming resources to enable multiple radio channels to be processed simultaneously.

10.5 Interpretability of Deep Learning Models

A further significant limitation is the poor interpretability of deep learning systems. Use cases driven by regulatory and compliancy demands require transparency—clients must understand why certain segments may indicate advertisements, political coverage, or prohibited content. Current CNNs and transformer models act as “black boxes,” making them less suitable for industries where stakeholders have high sensitivity. Future work should include explainable AI (XAI) techniques, including attention heatmaps, saliency maps, or concept bottleneck models, for interpretability. Hybrid frameworks that utilize interpretable rule-based cues along with deep embeddings may provide a better balance of interpretability, accuracy, and transparency.

10.6 Weakly-Supervised and Self-Supervised Learning

With annotated datasets being in short supply, weakly-supervised and self-supervised methods have emerged as promising alternatives. Whereas weak supervision involves utilizing high-level labels (e.g., “this clip has music”) without precise boundaries, self-supervised methods—such as contrastive predictive coding and masked prediction used in HuBERT and Wav2Vec 2.0—allow a model to learn rich representations from unlabeled audio clips. These methods significantly reduce annotation burden and increase generalization. Future work should pursue hybrid paradigms that incorporate weak supervision along with YOHO-style regression, which would allow a model to predict boundaries when only weak supervision is available.

10.7 Multimodal Fusion (Audio-Visual Integration)

Analysis of radio is increasingly associated with digital media streams that include both audio and video. The limitations of analysis focused on audio may hinder classification performance, particularly for the overlapping or ambiguous events present in radio broadcasts. However, a multimodal approach that integrates audio embeddings with visual cues (e.g., tracking a face, logo of advertisement, etc.) will help overcome the limitations of audio- based analysis. Transformer-based models for multimodal fusion integrate audio representation and visual representation to make decisions jointly. Applications of this multimodal approach have been used in speaker diaraization/lip tracking that features audio signals and advertisement discovery that uses logo recognition to enhance the audio. A major goal of this work moving forward is to expand the existing progress of multimodal methods for radio and online sources.

10.8 Hybrid Architectures and Emerging Paradigms

At last, hybrid architectural approaches, which merge valuable aspects of different paradigms, are expected to dominate future research. The CNN–Transformer hybrid model taps into CNN context to derive local feature extraction and transformer methods to grasp globally relevant context. Similarly, models that merge YOHO’s boundary regression with self-supervised transformer embeddings have the potential to yield models that are efficient while being semantically powerful. Emerging paradigms that will influence the future landscape of audio segmentation and classification during the next decade include federated learning (privacy-preserving distributed training), neural architecture search (NAS) as a model design approach, and edge AI deployment.

11 Conclusion

This review provides a comprehensive review of audio segmentation and classification techniques, focusing on their applications in radio broadcast analytics. We explored the evolution from rule-based methods and statistical machine learning models, such as HMMs and SVMs, to deep learning approaches leveraging convolutional, recurrent, and transformer-based architectures. Notably, we highlighted the YOHO (You Only Hear Once) algorithm, which reformulates segmentation as boundary regression, offering improved temporal precision, efficiency, and real-time applicability.

We also proposed a hybrid AI-based framework that integrates rule-based preprocessing, classical ML, deep learning, and YOHO for accurate and scalable radio analysis. The framework addresses challenges in segmentation, classification, and compliance monitoring, bridging the gap between research and practical implementation. The shift from handcrafted features like MFCCs to learned embeddings such as Wav2Vec and CLAP underscores the transition to data-driven methods, with deep learning frameworks offering improved robustness and scalability.

Despite advancements, challenges remain, such as noise robustness, overlapping acoustic events, and data scarcity. Future research directions include weakly-supervised learning, multimodal fusion, and hybrid models combining CNNs, transformers, and YOHO-style regression. Overall, this review highlights the potential of AI-driven approaches, particularly YOHO, to advance broadcast monitoring, indexing, and compliance as the demand for scalable audio content analysis grows.

Author Contributions Both authors contributed equally.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

References

1. Aggarwal S, G V, Selvakanmani S, Pant B, Kaur K, Verma A, Binegde GN (2022) Audio segmentation techniques and applications based on deep learning. Sci Program 2022(1):1–9. https://doi.org/10.1155/2022/7994191
2. Theodorou T, Mporas I, Fakotakis N (2014) An overview of automatic audio segmentation. IJITCS (IJITCS), vol. 6(11):1. https://doi.org/10.5815/ijitcs.2014.11.01
3. Boddapati V, Petef A, Rasmusson J, Lundberg L (2017) Classifying environmental sounds using image recognition networks. Proc Comp Sci 112:2048–2056. https://doi.org/10.1016/j.procs.2017.08.250
4. McLaren M, Ferrer L, Castan D, Lawson A (2016) The speakers in the wild (sitw) speaker recognition database (pp 818–822)
5. Typke R (2021) A survey of music information retrieval systems
6. Town SM, Bizley JK (2013) Neural and behavioral investigations into timbre perception. Front Syst Neurosci 7:88. https://doi.org/10.3389/fnsys.2013.00088
7. McAdams S (2019) The perceptual representation of timbre (pp 23–57)
8. Wattenberg M, Viégas FB, Hollenbach K (2007) Visualizing activity on wikipedia with chromograms (pp 272–287)
9. Schirmer A, Escoffier N, Cheng X, Feng Y, Penney TB (2016) Detecting temporal change in dynamic sounds: on the role of stimulus duration, speed, and emotion. Front Psychol 6:2055. https://doi.org/10.3389/fpsyg.2015.02055
10. Jeong I-Y, Lee K (2016) Learning temporal features using a deep neural network and its application to music genre classification (pp 434–440)
11. Singh N Speaker diarization: applications and challenges
12. Gandica Y, Aidos FSD, Carvalho J (2014) The dynamic nature of conflict in wikipedia. EPL 108(1):18003. https://doi.org/10.1209/0295-5075/108/18003
13. Graves A, Mohamed A-R, Hinton G (2013) Speech recognition with deep recurrent neural networks pp 6645–6649
14. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition pp 770–778
15. Vincent E, Virtanen T, Gannot S (2018) Audio source separation and speech enhancement
16. Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser Ł, Polosukhin I (2017) Attention is all you need. Adv Neural Inf Process Syst 30
17. Kumar A, Raj B (2016) Audio event detection using weakly labeled data pp 1038–1047
18. Takahashi N, Gygli M, Van Gool L (2018) Aenet: learning deep audio features for video analysis. IEEE Trans Multimedia 20(3):513–524. https://doi.org/10.1109/TMM.2017.2751969
19. Salamon J, Jacoby C, Bello JP (2014) A dataset and taxonomy for urban sound research pp 1041–1044
20. Goodfellow IJ, Pouget-Abadie J, Mirza M, Xu B, Warde-Farley D, Ozair S, Courville A, Bengio Y (2014) Generative adversarial nets. Adv. Neural Inf. Process. Syst 27
21. Adam KDBJ et al (2014) A method for stochastic optimization," arXiv preprint arXiv:1412.6980, vol. 1412(6)
22. Chung JS, Huh J, Nagrani A, Afouras T, Zisserman A (2020) Spot the conversation: speaker diarisation in the wild," arXiv preprint arXiv:2007.01216
23. Hershey JR, Chen Z, Roux JL, Watanabe S (2016) Deep clustering: discriminative embeddings for segmentation and separation (pp 31–35)
24. Kim JW, Salamon J, Li P, Bello JP (2018) Crepe: a convolutional representation for pitch estimation pp 161–165
25. Siegler MA, Jain U, Raj B, Stern RM (1997) Automatic segmentation, classification and clustering of broadcast news audio pp 97–99
26. Baevski A, Zhou Y, Mohamed A, Auli M (2020) wav2vec 2.0: a framework for self-supervised learning of speech representations. Adv Neural Inf Process Syst 33:12449–12460
27. Simonyan K, Zisserman A (2014) Very deep convolutional networks for large-scale image recognition," arXiv preprint arXiv:1409.1556
28. Purwins H, Li B, Virtanen T, Schlüter J, Chang S-Y, Sainath T (2019) Deep learning for audio signal processing. IEEE J Sel Top Signal Process 13(2):206–219. https://doi.org/10.1109/JSTSP.2019.2908700
29. Chollet F (2017) Xception: deep learning with depthwise separable convolutions pp 1251–1258
30. Schneider S, Baevski A, Collobert R, Auli M (2019) wav2vec: unsupervised pre-training for speech recognition," arXiv preprint arXiv:1904.05862
31. Zhu H, Luo M-D, Wang R, Zheng A-H, He R (2021) Deep audio-visual learning: a survey. Int J Autom Comput 18(3):351–376. https://doi.org/10.1007/s11633-021-1293-0
32. Bengio Y, Simard P, Frasconi P (1994) Learning long-term dependencies with gradient descent is difficult. IEEE Trans Neural Netw 5(2):157–166. https://doi.org/10.1109/72.279181
33. Li K, Song Y, Dai L-R, McLoughlin I, Fang X, Liu L (2023) Ast-sed: an effective sound event detection method based on audio spectrogram transformer, pp 1–5
34. Piczak KJ (2015) Esc: dataset for environmental sound classification pp 1015–1018

35. Adavanne S, Politis A, Nikunen J, Virtanen T (2019) Sound event localization and detection of overlapping sources using convolutional recurrent neural networks. IEEE J Sel Top Signal Process 13(1):34–48. https://doi.org/10.1109/JSTSP.2018.2885636
36. Kong Q, Xu Y, Sobieraj I, Wang W, Plumbley MD (2019) Sound event detection and time–frequency segmentation from weakly labelled data. IEEE/ACM Trans Audio Speech Lang Process 27(4):777–787. https://doi.org/10.1109/TASLP.2019.2895254
37. Schlüter J, Grill T (2015) Exploring data augmentation for improved singing voice detection with neural networks pp 121–126
38. Venkatesh S, Moffat D, Miranda ER (2022) You only hear once: a yolo-like algorithm for audio segmentation and sound event detection. Appl Sci 12(7):3293. https://doi.org/10.3390/app12073293
39. Devlin J, Chang M-W, Lee K, Toutanova K (2019) Bert: pre-training of deep bidirectional transformers for language understanding pp 4171–4186
40. Pascual S, Ravanelli M, Serra J, Bonafonte A, Bengio Y (2019) Learning problem-agnostic speech representations from multiple self-supervised tasks," arXiv preprint arXiv:1904.03416
41. Bredin H, Yin R, Coria JM, Gelly G, Korshunov P, Lavechin M, Fustes D, Titeux H, Bouaziz W, Gill M-P (2020) Pyannote. audio: neural building blocks for speaker diarization pp 7124–7128
42. Ravanelli M, Parcollet T, Plantinga P, Rouhe A, Cornell S, Lugosch L, Subakan C, Dawalatabad N, Heba A, Zhong J et al (2021) Speechbrain: a general-purpose speech toolkit," arXiv preprint arXiv:2106.04624
43. Zaman K, Sah M, Direkoglu C, Unoki M (2023) A survey of audio classification using deep learning. IEEE Access. 11:106620–106649. https://doi.org/10.1109/ACCESS.2023.3318015
44. Zaman K, Li K, Sah M, Direkoglu C, Okada S, Unoki M (2025) Transformers and audio detection tasks: an overview. Digit Signal Process 158:104956. https://doi.org/10.1016/j.dsp.2024.104956
45. LeCun Y, Bengio Y, Hinton G (2015) Deep learning. Nature 521(7553):436–444. https://doi.org/10.1038/nature14539
46. Unoki M, Li K, Chaiwongyen A, Nguyen Q-H, Zaman K (2024) Deepfake speech detection: approaches from acoustic features related to auditory perception to deep neural networks. IEICE Trans Inf Syst
47. Zaman K, Samiul IJ, Sah M, Direkoglu C, Okada S, Unoki M (2024) Hybrid transformer architectures with diverse audio features for deepfake speech classification. IEEE Access. 12:149221–149237. https://doi.org/10.1109/ACCESS.2024.3478731
48. Boersma P (2001) Praat, a system for doing phonetics by computer. Glot Int 5(9):341–345
49. Anguera X, Bozonnet S, Evans N, Fredouille C, Friedland G, Vinyals O (2012) Speaker diarization: a review of recent research. IEEE Trans Audio Speech Lang Process 20(2):356–370. https://doi.org/10.1109/TASL.2011.2125954
50. Ahmed I, Khalil A, Ahmed I, Frnda J (2022) Sparse signal representation, sampling, and recovery in compressive sensing frameworks. IEEE Access. 10:85002–85018. https://doi.org/10.1109/ACCESS.2022.3197594
51. Li K, Zaman K, Li X, Akagi M, Dang J, Unoki M (2025) Machine anomalous sound detection using spectral-temporal modulation representations derived from machine-specific filterbanks. IEEE Trans Audio Speech Lang Process 33:2059–2073. https://doi.org/10.1109/TASLPRO.2025.3570956
52. Zaman K, Li K, Samiul IJ, Uezu Y, Kidani S, Unoki M (2025) Ability of human auditory perception to distinguish human-imitated speech. IEEE Access. 13:6225–6236. https://doi.org/10.1109/ACCESS.2025.3526631
53. Ahmed I, Ahmad N, Ali H, Ahmad G (2012) The development of isolated words pashto automatic speech recognition system. In 18th International conference on automation and computing (ICAC), IEEE, pp 1–4
54. Ahmed I, Ali H, Ahmad N, Ahmad G (2012) The development of isolated words corpus of pashto for the automatic speech recognition research. In 2012 International conference of robotics and artificial intelligence, IEEE, pp 139–143
55. Ahmed I, Irfan MA, Iqbal A, Khalil A, Siddiqui SI (2023) Efficient feature extraction and classification for the development of pashto speech recognition system. Multimed Tools Appl 83(18):54081–54096. https://doi.org/10.1007/s11042-023-17684-w
56. Ahmed I, Khan A (2022) Genetic algorithm based framework for optimized sensing matrix design in compressed sensing. Multimed Tools Appl 81(27):39077–39102. https://doi.org/10.1007/s11042-022-12894-0
57. Ahmed I, Khan A (2023) Learning based speech compressive subsampling. Multimed Tools Appl 82(10):15327–15343. https://doi.org/10.1007/s11042-022-14003-7
58. Ahmed I, Khan A, Ahmad N, N, Ali H (2020) Speech signal recovery using block sparse bayesian learning. Arab J Sci Eng 45(3):1567–1579. https://doi.org/10.1007/s13369-019-04080-6
59. Ahmed I, Khan A, Khan A, Mujahid K, Khan N (2021) Efficient measurement matrix for speech compressive sampling. Multimed Tools Appl 80(13):20327–20343. https://doi.org/10.1007/s11042-021-10657-x
60. Ahmed I, Wuttisittikulkij L, Khan A, Iqbal A (2024) The optimally designed deep autoencoder-based compressive sensing framework for 1d & 2d signals. IEEE Access. 12:150520–150539. https://doi.org/10.1109/ACCESS.2024.3472044
61. Veeramanickam M, Ingavale A, Khullar V, Tirth SS, Pandey B, Singh HP (2024) Identification of sounds using deep learning with mfcc features extraction pp 60–64
62. Telmem M, Laaidi N, Satori H (2025) The impact of mfcc, spectrogram, and mel-spectrogram on deep learning models for amazigh speech recognition system. Int J Speech Technol 28(1):299–312. https://doi.org/10.1007/s10772-025-10183-3
63. Zhang T, Liu Y (2025) Research on acoustic scene classification based on time–frequency–wavelet fusion network. Sensors 25(13):3930. https://doi.org/10.3390/s25133930
64. Nurmemet Y, Sun L, Li X, Liu Q, Wang Z (2024) Wav2vec-ad: acoustic unit discovery module-integrated, self-supervised contrastive pre-training approach for speech recognition. J Educ Chang Shanghai Jiaotong Univ (Sci) 1–9
65. Tseng W-C, Harwath D (2025) Codec2vec: self-supervised speech representation learning using neural speech codecs. https://doi.org/arXivpreprintarXiv:2511.16639
66. Gong Y, Chung Y-A, Glass J (2021) Ast: audio spectrogram transformer
67. Erol MH, Senocak A, Feng J, Chung JS (2024) Audio mamba: bidirectional state space model for audio representation learning. IEEE Signal Process Lett 31:2975–2979. https://doi.org/10.1109/LSP.2024.3483009
68. Liang D, Li X (2025) Ls-eend: long-form streaming end-to-end neural diarization with online attractor extraction. IEEE Trans Audio Speech Lang Process 33:3568–3581. https://doi.org/10.1109/TASLPRO.2025.3597446
69. Plaquet A, Tawara N, Delcroix M, Horiguchi S, Ando A, Araki S (2025) Mamba-based segmentation model for speaker diarization (pp 1–5)
70. Rybicka M, Villalba J, Thebaud T, Dehak N, Kowalczyk K (2024) End-to-end neural speaker diarization with non-autoregressive attractors. In IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol 32. pp 3960–3973

71. Ahmed M, Alharbey R, Daud A, Khan HU, Nawal J, Arshad G (2025) An enhanced deep learning approach for speaker diarization using titanet, marbelnet and time delay network. Sci Rep 15(1):24501. https://doi.org/10.1038/s41598-025-09385-1
72. Yasuda S, Nguyen TNT, Komatsu T, Kinoshita K, Araki S et al (2025) Description and discussion on DCASE 2025 challenge task 4," arXiv preprint arXiv:2506.10676
73. Nguyen BT, Yasuda M, Takeuchi D, Niizumi D, Ohishi Y, Harada N (2025) Baseline systems and evaluation metrics for spatial semantic segmentation of sound scenes. IEEE, pp 266–270
74. Cornell S, Ebbers J, Douwes C, Martín-Morató I, Harju M, Mesaros A, Serizel R (2024) Dcase 2024 task 4: sound event detection with heterogeneous data and missing labels," arXiv preprint arXiv:2406.08056
75. Martín-Morató M, Serra X (2025) AudioSet-tools: a python framework for taxonomy-aware AudioSet curation. J Audio, Speech, Music Process